Solar Energy 137 (2016) 25–45

Contents lists available at ScienceDirect

Solar Energy

journal homepage: www.elsevier.com/locate/solener

Review

Global advancement of cooling technologies for PV systems: A review

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ARTICLE INFO

Article history:
Received 27 April 2016
Received in revised form 1 July 2016
Accepted 6 July 2016
Available online 4 August 2016

Keywords:
Solar energy
Photovoltaic
Passive cooling
Active cooling
Heat transfer
Temperature

ABSTRACT

Photovoltaic (PV) systems operate in a paradox; sunlight is the essential input to generate electricity with PV, but they suffer a digression in performance as the operating temperature goes higher. This work is a comprehensive compilation and review of the latest literature regarding research works rendered to achieve improved efficiency through appropriate cooling systems. Most of the research goals were two-fold, that is to enhance the efficiency of the solar PV systems and to ensure a longer life at the same time. The passive cooling systems are found to achieve a reduction in PV module temperature in the range of 6–20 °C with an improvement in electrical efficiency up to 15.5% maximum. On the other side, active cooling systems' performance are better, as may expected, with a reduction in PV module temperature as high as 30 °C with an improvement in electrical efficiency up to 22% maximum along with additional thermal energy output with efficiency reaching as high as 60%. Based on the wide-ranging review, it may be predicted that with the swelling growth of solar PV electricity worldwide, the compatible cooling system is becoming obligatory in order to ensure better energy harvest and utilization.

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http://dx.doi.org/10.1016/j.solener.2016.07.010
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Nomenclature

b	width of PVT collector (m)
A	area
C_p	specific heat capacity of water (J/kg K)
D	width of absorber plate on a flow pipe, diameter of flow pipe (m)
$\dot{E}$	power (W)
$\dot{E}x$	exergy rate
F_R	heat removal factor
F'	collector efficiency factor
G	solar radiation intensity (W/m^2)
h_{p1}	penalty factor due to the presence of solar cells material, glass, and EVA
I	current (A)
L	dimensions of solar module, the length of PVT water collector (m)
$\dot{m}$	mass flow rate of water (kg/s)
P	pressure (Pa)
$\dot{Q}$	heat transfer rate (W)
T	temperature (K)
$\overline{T}$	average temperature (K)
U_L	overall heat transfer coefficient from the PVT collector to the environment (W/m^2 K)
U_t	overall heat transfer coefficient from solar cells to the ambient through glass cover (W/m^2 K)
U_p	overall heat transfer coefficient from solar cells to absorber plate (W/m^2 K)
U_{tp}	overall heat transfer coefficient from glass to absorber plate through solar cells (W/m^2 K)
U_{tf}	overall heat transfer coefficient from glass to agent fluid through solar cells (W/m^2 K)
V	voltage (V)

Greek symbols

α	absorptivity, current temperature coefficient (mA/°C)
$(\alpha\tau)_{eff}$	product of effective absorptivity and emissivity
β	packing factor, voltage temperature coefficient (V/°C), the temperature coefficient of electrical efficiency (1/°C)
Δ	difference in current, pressure, temperature, voltage
ε	semiconductor bandgap energy (eV)
η	efficiency (%)
ρ	density (kg/m^3)
τ	transmissivity

Subscripts

a	ambient
c	cell
des	destroyed
el	electrical
ex	exergy
f	fluid flow
g	glass
in	inlet
int	internal
$loss$	loss
mp	maximum power point
p	absorber plate, pump
PVT	PVT collector
oc	open circuit
opt	optimum
out	outlet
Q	heat transfer
ref	reference
s	series, sun
sc	short circuit
t	top
th	thermal

1. Introduction

Secure, unswerving, accessible and clean energy supply is the key to sustainable economic growth. Reducing the carbon footprint while fulfilling growing energy demand is major concern nowadays. Global energy demands are growing fast and the worldwide energy consumption in the decades of 2010–2030 will be increased by 33% (Hasanuzzaman et al., 2012; Ahmed et al., 2013). The proven and predicted energy sources are not sustainable and only renewable sources like solar energy can be the solution for sustainable power supply. At present about 81% energy comes from mineral sources (oil 32.4%, natural gas 21.4%, and coal 27.3%) while the share of renewable energy is only 13% and nuclear power 5.7% (Borges Neto et al., 2010; Hosenuzzaman et al., 2015). GHG (greenhouse gas) emission has grown at a rate of 2.2% per annum in the last decennia of 2000–2010 whereas this rate was as low as 1.3% for the previous three decades from 1970 to 2000 (Kalaugher, 2014). As a result, the volume of CO_2 and other harmful greenhouse gases is increasing steeply in the environment and worsening the climate state further. Renewable energy applications are the most prospective solution to this vulnerable situation. Solar photovoltaic (PV) electricity generation, among the renewable energy technologies, is one of the most potential options to encounter the future energy predicament. Being a clean source of energy, the PV system has a great potential, especially, in the tropical area.

Solar PV panels are increasingly used all over the world due to their capability to work under diffuse radiation. So, it is important to know how PV panels respond to different climatic conditions. In practice only 15–20% of the solar irradiation can be transformed into electricity, the rest being wasted as heat (Teo et al., 2012). PV module efficiency decreases at a rate of approximately 0.40–0.65% with a one-degree increment of module temperature (Rahman et al., 2015; Shan et al., 2014). PV temperature can reach as high as 80 °C particularly in hot arid regions (Reddy et al., 2015). So, continuous efforts are being given to improving the efficiency of the photovoltaic cells by controlling the cell temperature. In the tropical regions, the temperature may go out of the operating ranges (Koehl et al., 2012) which consequence thermal degradation and reduction in efficiency. Experimental studies are continuously going on to find a cost effective solution for the performance improvements of solar photovoltaic modules. Zhao et al. (2011) carried out investigation by attaching fins and providing a forced convection. The results showed the exergetic efficiency increases from 12% to 22%. Pandey et al. (2013) performed a performance evaluation and parametric study of a multi-crystalline solar photovoltaic module using energy and exergy analysis for at a typical climatic condition of north India. The authors found the best energy efficiency for the month of December and the least exergy efficiency was found in July. Pearce (2008) performed experiments on another study funded by Heriot–Watt Dubai campus tested the SPC (panel cooling equipment from S&P Coil Products Limited, UK) for heat removal from the PV panels and found the fin to be effective to bring back the panel temperature to the operating range again. This passive heat transfer phenomenon removed heat by

natural convection. Analysis using CFD (Computational Fluid Dynamics) technique has been carried out for copper heat pipe with fins where water was used as the working fluid. The optimum temperature cooled under the hot and arid environmental conditions was found in the range of solar cells operating temperature of 30 °C. These studies encourage an ongoing investigation in this direction finally to find out an optimum solution to create a cooling system (Hughes et al., 2011; Zuser and Rechberger, 2011).

Researchers proposed hybrid photovoltaic and thermal (PVT) collectors that improve system performance. A hybrid PVT system simultaneously provides electrical and thermal energy, thereby improving the overall system efficiency. PVT makes an extended use of the solar energy by concomitant electrical and thermal energy. This is achieved by using a coolant, such as air, water or any refrigerant to take away the excess heat that would otherwise raise the PV cell temperature. The heat thus extracted can be used for heating applications. According to Pandey et al. concentrated photovoltaics (CPV) and photovoltaic-thermal (PVT) are technically most sound and feasible technologies to address future energy challenges (Pandey et al., 2016).

There are some comprehensive reviews on different thermal management techniques of PVT. Bahaidarah et al., in a recent review, highlighted the importance of uniform cooling of PV modules and showed its comparative merits over non-uniform cooling (Bahaidarah et al., 2016). Ma et al. carried out an all-inclusive appreciation on the use of phase change materials (PCM) for thermal regulation and electrical efficiency improvement of PV modules (Ma et al., 2015). Browne et al. also made such an overview on the usage of PCM in temperature control of PVT (Browne et al., 2015). Sahay et al. presented an appraisal on the cooling technologies for PV panels with special treatment of a ground-coupled central panel cooling system (GC-CPCS) (Sahay et al., 2015). Du et al. reviewed the various cooling technologies including passive and active methods those are in being applied in PV installations (Du et al., 2013).

As solar energy is recently considered as a sustainable alternative to the conventional ones, so comprehensive research works including regular review and follow-up of the ongoing researches is indispensable. This review pursues the effect of temperature on PV systems and the progress attained so far in solving the problem of efficiency drop due to temperature rise. The paper will be helpful for the researchers, manufacturers, decision makers as well as consumers to be acquainted with current and updated PV technologies and get better performance from the systems.

2. Thermal modelfor PV modules

Before addressing the consequence of elevated temperatures of the solar cell on their energy conversion efficiency, it may be relevant to have a look into the thermal models that have addressed the energetic as well as exergetic aspects of a solar photovoltaic. In literature, there are several such models as proposed by different researchers. A comprehensive thermal model for photovoltaic thermal (PVT) hybrid system taking both energy and exergy analysis into account has been compiled by Yazdanpanahi et al. (2015) in light of the some previous relevant works and the outcomes obtained from energy analysis are listed as below:

Solar cell and absorber plate temperatures have been calculated by considering thermal balance among different layers of the solar PV module. The results are as below,

Solar cell temperature (Tiwari and Dubey, 2010),

$$T_c = \frac{\tau_g \beta_c (\alpha_c - \eta_{el}) G + U_t T_a + U_p T_p}{U_t + U_p} \tag{1}$$

Absorber plate temperature (Tiwari and Dubey, 2010),

$$T_p = \frac{[\tau_g^2 \alpha_p (1-\beta_c) + \tau_g \beta_c (\alpha_c - \eta_{el}) h_{p1}] G + U_{tp} T_a + U_f T_f}{U_{tp} + U_f} \tag{2}$$

where $h_{p1} = \frac{U_p}{U_t + U_p}$ and $U_{tp} = \frac{U_t U_p}{U_t + U_p}$.

Water outlet temperature of the PVT is a function of ambient and water inlet temperatures, it also depends on the value of heat removal factor, i.e., the rate at which heat is being taken away from the PV module (Tiwari and Dubey, 2010),

$$T_{f,out} = \left(T_a - T_{f,in} + \frac{(\alpha\tau)_{eff} G}{U_L}\right)\left(\frac{F_R U_L b L}{\dot{m} C_p}\right) + T_{f,in} \tag{3}$$

Heat removal factor is calculated as

$$F_R = \frac{\dot{m} C_p}{U_L b L}\left[1 - \exp\left(\frac{-bF' U_L L}{\dot{m} C_p}\right)\right] \tag{4}$$

Rate of useful thermal energy that may be harvested from the PVT is computed by the amount of heat content of the coolant fluid which depends on its specific heat capacity, mass flow rate and the temperature difference between the inlet and outlet (Tiwari and Dubey, 2010),

$$\dot{Q}_u = \dot{m} C_p (T_{f,out} - T_{f,in}) = F_R b L [(\alpha\tau)_{eff} G - U_L (T_{f,in} - T_a)] \tag{5}$$

Thermal efficiency of the PVT water collector (Tiwari and Dubey, 2010),

$$\eta_{th} = \frac{Q_u}{G A_{PVT}} = F_R \left[(\alpha\tau)_{eff} - \frac{U_L (T_{f,in} - T_a)}{G}\right] \tag{6}$$

Electrical power consumption by pump to maintain water circulation in the PVT water collectors calculated as (De Soto et al., 2006; Townsend et al., 1989),

$$\dot{E}_p = \frac{\dot{m} \Delta P}{\rho \eta_p} \tag{7}$$

In calculating the electrical efficiency of the PVT, the power consumed by the pump must be taken into account. So, PVT electrical efficiency is calculated as below (De Soto et al., 2006; Townsend et al., 1989),

$$\eta_{el} = \frac{V_{mp} I_{mp} - \dot{E}_p}{G A_{PVT}} \tag{8}$$

The outcomes obtained from exergy analysis are listed as below:

The incident solar energy is not entirely utilized as effective input to the PV module. The net input exergy rate of PVT water collector is a function of both the sun and the ambient temperatures (Petela, 2003, 2008),

$$\sum \dot{E}x_{in,net} = \dot{E}x_{Q,s} = G A_{PVT}\left[1 - \frac{4}{3}\left(\frac{T_a}{T_s}\right) + \frac{1}{3}\left(\frac{T_a}{T_s}\right)^4\right] \tag{9}$$

The net output exergy rate of PVT consists of two components, one is thermal exergy and the other one is electrical exergy (Tiwari et al., 2009),

$$\sum \dot{E}x_{in,net} = \dot{E}x_{th} + \dot{E}x_{el} \tag{10}$$

The thermal exergy includes the exergy changes of water flow in PVT collector (Tiwari et al., 2009),

$$\dot{E}x_{th} = \dot{Q}_u \left(1 - \frac{T_a}{T_{f,out}}\right) \tag{11}$$

The electrical exergy is obtained by excluding the energy consumption of the pump or fan from the electrical power,

$$\dot{E}x_{el} = \dot{E}_{el} - \dot{E}_p \tag{12}$$

Electrical power is given by an empirical correlation (Chow et al., 2009; Dubey and Tiwari, 2008; Fudholi et al., 2014; Mishra and Tiwari, 2013; Tiwari et al., 2009; Tiwari and Sodha, 2006),

$$\dot{E}_{el} = \eta_{el} G A_{PVT} = \eta_{el,ref}[1 - \beta_{ref}(T_c - T_{a,ref})] G A_{PVT} \quad (13)$$

The exergy efficiency of the PVT water collector is the ratio of the total exergy output to the exergy input,

$$\eta_{ex} = \frac{\dot{Q}_u\left(1 - \frac{T_a}{T_{f,out}}\right) + \eta_{el,ref}[1 - \beta_{ref}(T_c - T_{a,ref})] G A_{PVT} - \dot{E}_p}{G A_{PVT}\left[1 - \frac{4}{3}\left(\frac{T_a}{T_s}\right) + \frac{1}{3}\left(\frac{T_a}{T_s}\right)^4\right]} \quad (14)$$

Exergy losses rate of the PVT water collector comprises of six components; the first is that caused by optical losses in PVT collector surface (Faramarz et al., 2010; Sarhaddi et al., 2010a),

$$\dot{E}x_{loss,opt} = G A_{PVT}\left[1 - \frac{4}{3}\left(\frac{T_a}{T_s}\right) + \frac{1}{3}\left(\frac{T_a}{T_s}\right)^4\right] - (\alpha\tau)_{eff} G A_{PVT}\left[1 - \frac{4}{3}\left(\frac{T_a}{T_s}\right)\Delta + \frac{1}{3}\left(\frac{T_a}{T_s}\right)^4\right] \quad (15)$$

The second font in exergy destruction is owing to the temperature difference between the sun and PVT collector surface (Faramarz et al., 2010; Sarhaddi et al., 2010a),

$$\dot{E}x_{des,\Delta T_{s,PVT}} = (\alpha\tau)_{eff} G A_{PVT}\left[1 - \frac{4}{3}\left(\frac{T_a}{T_s}\right) + \frac{1}{3}\left(\frac{T_a}{T_s}\right)^4\right] - (\alpha\tau)_{eff} G A_{PVT}\left(1 - \frac{T_a}{T_c}\right) \quad (16)$$

The third term in exergy destruction is owing to heat transfer from PVT surface to the working fluid at a finite temperature difference (Yazdanpanahi et al., 2015),

$$\dot{E}x_{des,\Delta T_{PVT,f}} = (\alpha\tau)_{eff} G A_{PVT}\left(1 - \frac{T_a}{T_c}\right) - \left[\dot{Q}_{loss}\left(1 - \frac{T_a}{T_c}\right) + \dot{Q}_u\left(1 - \frac{T_a}{T_{f,out}}\right) + V_{oc} I_{sc}\right] \quad (17)$$

where $\dot{Q}_{loss}$ is the loss of heat rate from the PVT to surrounding and given by (Joshi et al., 2009; Sarhaddi et al., 2011),

$$\dot{Q}_{loss} = U_L A_{PVT}(T_c - T_a) \quad (18)$$

The fourth part of exergy destruction is caused by heat loss from PVT system to surrounding (Faramarz et al., 2010; Joshi et al., 2009; Sarhaddi et al., 2011),

$$\dot{E}x_{loss,opt} = \dot{Q}_{loss}\left(1 - \frac{T_a}{T_c}\right) \quad (19)$$

The fifth term is exergy destruction due to pressure drop in flow pipes (Kotas, 1995; Wong, 2011),

$$\dot{E}x_{des,\Delta P} = \frac{T_a \dot{m} \Delta P}{\rho \overline{T}_f} \quad (20)$$

The sixth term is electrical exergy destruction rate (Faramarz et al., 2010; Joshi et al., 2009; Sarhaddi et al., 2010a),

$$\dot{E}x_{des,el} = V_{oc} I_{sc} - (V_{mp} I_{mp} - \dot{E}_p) \quad (21)$$

In this connection, some other important correlations as presented by Teo et al. (2012) are mention worthy. These are compiled as follows.

The total absorbed energy by the PV cell is (Cox and Raghuraman, 1985),

$$E_c = p \alpha_c \tau_g G \quad (22)$$

where p is the ratio of the area of the solar cell to the area of the blank absorber and known as a packing factor, α_c is the cell absorptivity, τ_g is the transmitted irradiation through the front glass, G is the solar radiation.

The electricity produced by the PV cell is (Cox and Raghuraman, 1985),

$$E_{ce} = \eta_e p \tau_g G \quad (23)$$

The cell efficiency is given by (Cox and Raghuraman, 1985),

$$\eta_e = \eta_o[1 - \beta(T_c - T_a)] \quad (24)$$

where β temperature coefficient and η_o is the nominal electrical efficiency under the standard condition and is given by $\eta_o = \frac{V_{mp} I_{mp}}{GA}$.

The heat released by the PV cell is (Cox and Raghuraman, 1985),

$$E_{ct} = (1 - \eta_e/\alpha_c) p \alpha_c \tau_g G \quad (25)$$

The solar energy absorption rate by Tedler back sheet after transmission from EVA (polymer encapsulation of solar cell) is (Cox and Raghuraman, 1985),

$$E_T = \tau_g (1 - p) \alpha_T \quad (26)$$

where α_T is the Tedler absorptivity.

3. Impact of temperature on PV performance

It is now established that temperature degenerates the performance PV devices. PV modules require sunlight for electricity generation, but the heat from the sun deteriorates their conversion ability. This enigma has challenged the industry for years.

The utmost affected parameter due to temperature rise in a solar cell or PV module is its open-circuit voltage. Raise in temperature increase the resistance in the circuit which reprehends the speed of the electrons. The properties of solar cell materials are also affected by the temperature. Therefore, a PV system should be contrived not only according to the ambient temperature trend but also with a thorough knowledge of the materials used in the PV panel.

Temperature coefficient, i.e., drop in voltage per unit rise in temperature, of a particular material characterizes the temperature dependence of the PV module performance. For polycrystalline PV

Table 1
Performance of various types of cells and their temperature coefficient (Seng, 2010).

Type	STC performance (%)	Temperature coefficient
Monocrystalline-Si (m-Si)	12.5–15	−0.4 to −0.5
Polycrystalline-Si (p-Si)	11–14	−0.4 to −0.5
Amorphous-Si (a-Si)	11–13	−0.35 to −0.38
CIGS	10–13	−0.32 to −0.36
CdTe	9–12	−0.25

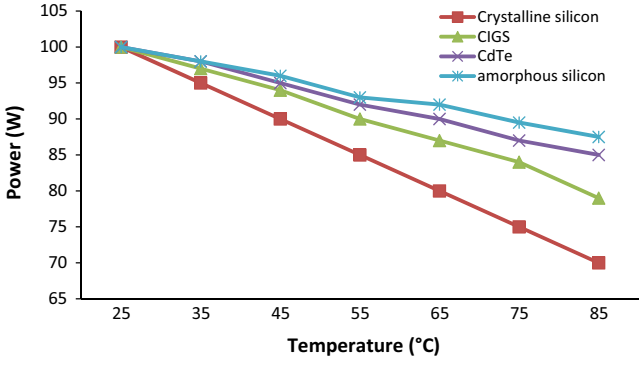


Fig. 1. Effect of temperature on different PV materials (Seng, 2010).

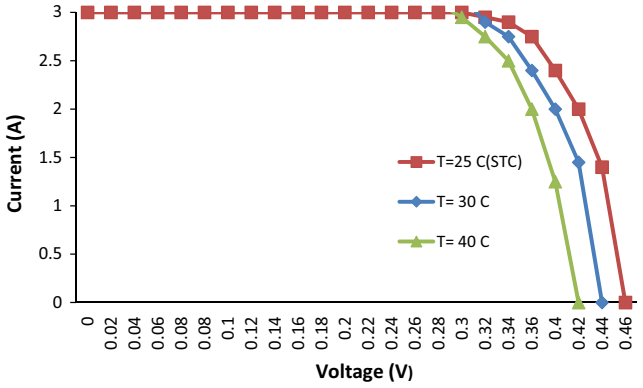


Fig. 2. *I–V* physiognomies of a multi-crystalline Si cell (Krauter and Ochs, 2003).

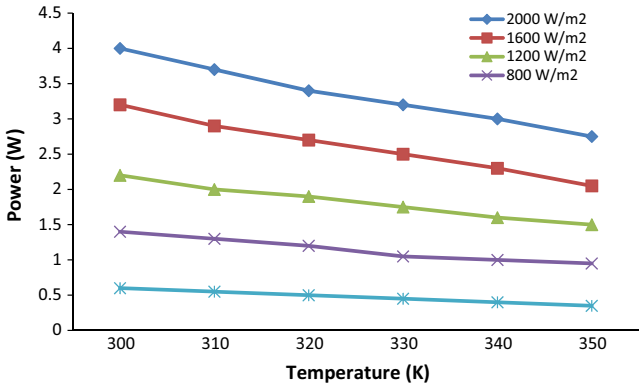


Fig. 3. Cell temperature variation with different irradiance (Najafi and Woodbury, 2013).

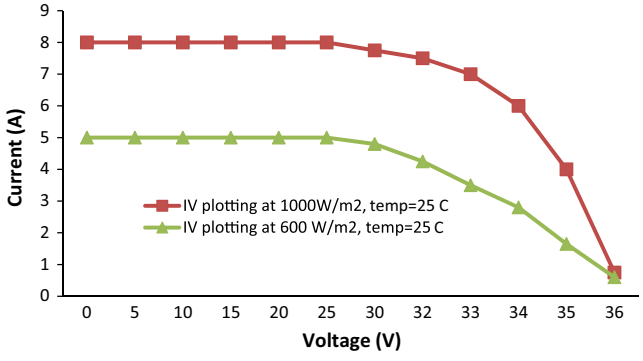


Fig. 4. *I–V* characteristic curve of multi-crystalline Si-PV modules (Jiang et al., 2012).

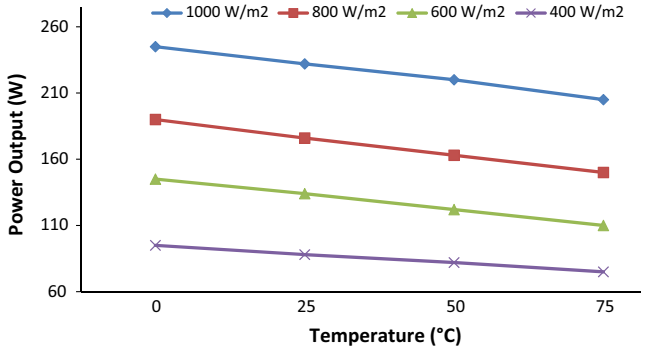


Fig. 5. The maximum power output of the mono-crystalline Si-PV modules (Jiang et al., 2012).

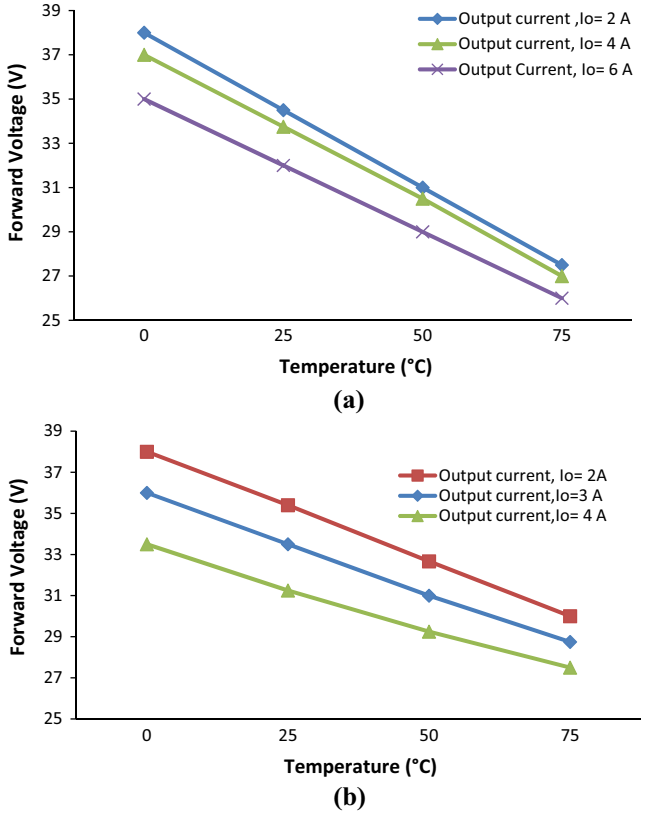


Fig. 6. Forward voltage in mono-crystalline silicon PV modules (a) at 1000 W/m^2 (b) at 600 W/m^2 (Jiang et al., 2012).

modules, the voltage drop is 0.12 V for each degree rise in cell temperature. So the temperature coefficient is 0.12 V/°C for polycrystalline cells. Table 1 shows the standard testing condition performance and negative temperature coefficients of different types of cells (Seng, 2010). The effects of temperature rise on the power output of a PV cell for different manufacturing materials are also clarified in Fig. 1.

Apart from the negative temperature coefficients, there are several other factors which affect the performance of the photovoltaic modules. These factors are called key performance indicators (KPI). Some of the KPIs are losses in connection and circuitry, losses due to climatic conditions and losses in cabling, shading loss etc. Another important parameter is the tracking of maximum power point. These KPIs set the outdoor performance of the system.

Agroui tested the behavior of multi-crystalline PV modules under varied conditions (Agroui, 2012). The tests were carried out in the temperature range of 61–75 °C and the irradiance range 780–1250 W/m^2. The results indicate that the influence of a combination of high working temperature and high irradiance suppressed the free carrier loss mechanisms in the module. Under these operating conditions, the modules lost about 35% efficiency and 18% of the maximum output power compared with the results under standard testing conditions. It is also noted that the loss in these parameters is nonlinear with the working temperature and irradiance.

The *I–V* characteristics of solar cells under increased temperatures were studied by the researchers decades ago. The findings of Krauter and Ochs (Krauter and Ochs, 2003), as illustrated in Fig. 2, gives a general idea regarding the response of electrical

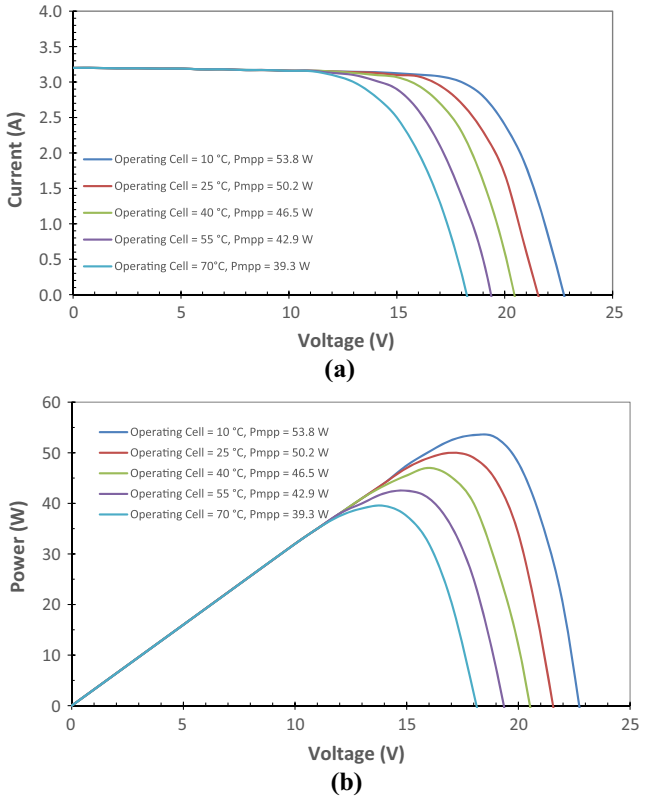


Fig. 7. Features of (a) output *I–V*, (b) output *P–V* of the PV module at various temperatures (Fesharaki et al., 2011).

current flow as a function of voltage change for a typical multi-crystalline silicon solar cell at diverse temperatures.

The maximum power generation points as depicted by Najafi and Woodbury (2013) are presented in Fig. 3. It may be observed that rated power yield of the cell under the standard testing condition (STC) will give a lower output below this irradiance level but unexpectedly it is giving lower output at higher irradiance levels beyond the STC. This is may be due to the increased temperature of the PV module at higher solar irradiations.

Jiang et al. (2012) experimented on the PV modules under a wide range of temperature of 0–75 °C and under four irradiation levels with intensities as 400, 600, 800, and 1000 W/m^2. Two of the outcomes of interest are shown in Figs. 4–7.

The experimental results reveal that the changes in the temperature of the PV cells bring changes in the level of power output. Under the same irradiation level different level of power are induced in the cells under varying temperatures, e.g., at the standard testing condition of 25 °C and 1000 W/m^2, the power output is about 235 W whereas at the irradiation level the power output drops to about 210 W when the temperature is increased to 75 °C. Fig. 6(a) and (b) shows that under the same irradiation level different current and voltages are induced in the cells under varying temperatures. The declining trend in forward voltage with increasing temperature demonstrates the fact that the voltage and power output decreases with increasing temperature which, according to the author, is due to the increase of junction temperature. Although it is very complex to analyze the actual mechanism, but theoretically it is confirmed that short-circuit current I_{sc} is amplified dramatically with the rise irradiation intensity. This may be due to the increase in the minority carrier concentration.

The effect of temperature on photovoltaic cell efficiency has been investigated by Fesharaki et al. (2011). The effect of temperature on output *I–V* and *P–V* features of the PV module as obtained by the above investigators are presented below.

Effect of temperature on PV module's *I–V* characteristics at constant radiation is shown in Fig. 7(a). As the temperature decreases PV current decrease marginally but the voltage decreases very noticeably. On the other hand, as can be seen from Fig. 7(b), the module power output increases with a decline in temperature.

4. Passive cooling technologies for solar PV systems

A number of cooling methods have been attempted and/or proposed by researchers for PV modules. PV module cooling may be rendered actively or passively. Active cooling systems are functioned by fans or pumps which require external power input while passive systems require no additional power to operate. PV array installed on the roof allow natural convection on the rear of the panel heat or a white color roof that prevents additional heat gain of the panel; these methods allow passive cooling of the PV. But this way is relatively inefficient and slow in terms of cooling attainment. Active cooling system, on the other hand, is very much effective especially in certain situations where the ambient temperature itself is very high such as in solar power plants installed in deserts or in the cases where enhanced performance is a requirement. Active cooling may serve an additional purpose like domestic water heating. This chapter compiles recent works and achievements on PV cooling with a special focus on some innovative passive cooling techniques.

Natural air circulation over and under the PV panel is perhaps the most facile and natural way of passive cooling. But, this is suitable mainly for stand-alone systems. In addition, most of the passive cooling methods are less effective yet time-consuming process. This section starts with some of the most recent and innovative approaches to achieving better performance with the passive cooling method.

4.1. Recent innovative passive cooling models

The effort to cool solar panels passively has got some of the most novel developments in recent years. Researchers prefer this cooling technique as it does not require additional power to operate.

A solar energy group at Stanford University very recently (2014) developed a colorless and lucid silicon coat that escalates the efficiency of solar cells by keeping the cell temperature low (Zhu et al., 2014). The transparent coating first collects and then radiates heat directly into space, without intervening with approaching photons. The authors report that application of radiative cooling in combination with the use of sunlight will open new possibilities to employ the coolness of the universe and improve energy yielding performance of the power systems. The silica (SiO_2) solar panel coating devised by the researchers is successfully making use of space as the largest of heatsinks. It does so by collecting and then radiating heat as infrared electromagnetic waves, which can easily travel through the atmosphere, out into space. The coating is transparent, so it won't interfere with the solar cell's light collecting ability, and improves on the heat dissipation of the silicon found in most cells. Data analysis revealed that the overlay was able to cool the underlying solar absorber by up to 13 °C (23 °F) which would confer an improvement in overall cell efficiency by one percent.

Another creative technique for cooling the PV cell has been introduced by Ebrahimi et al. (2015) that made the use of the natural vapor as heat transfer fluid. The experimental setup is presented in Fig. 8. The scheme devised by the investigators comprises of natural vapor simulator, solar simulator and solar cell along with the temperature and *V–I* measuring system.

The authors suggest that actual installation of panels could be placed on the rivers and canals where natural evaporation takes

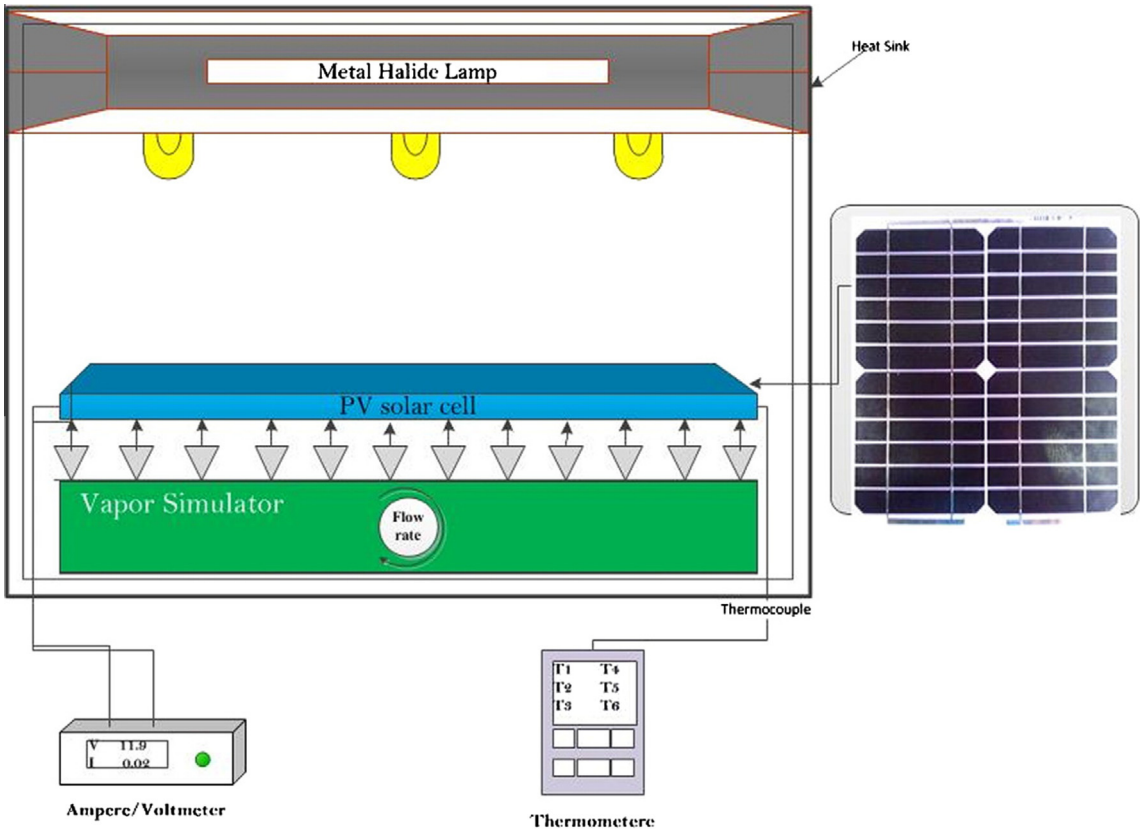


Fig. 8. Schematic of the natural vapor cooled solar panel (Ebrahimi et al., 2015).

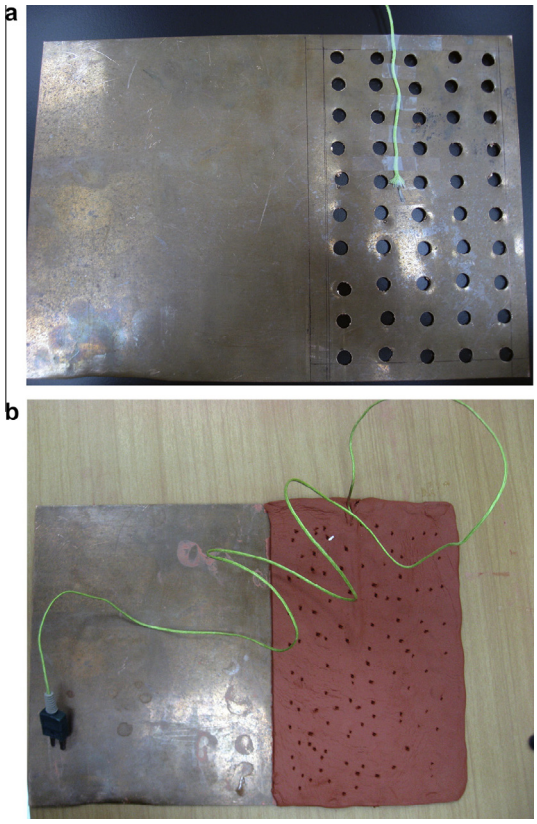


Fig. 9. (a) Copper sheet used for cooling simulation, (b) clay covered copper sheet with thermocouples (Alami, 2014).

place. This mode of cooling does not require any extra energy. Increasing flow rate of natural vapor augments the heat transfer rate from PV module, thus improving electrical efficiency. At low ambient temperature, the average cell temperature achieved was 48.3 °C with lower flow rates and 39.3 °C with high flow rates of natural vapor. According to the authors, maximum coverage of natural vapor in the backside of the PV results in the increase in 7.3% rise in power output.

Effect of evaporative cooling in alleviating the surplus cell temperature and increase the efficiency of photovoltaic modules has been investigated by Alami (2014). Simulated studies were done with copper sheet covered by synthetic clay to permit indirect evaporation and thereby cooling effect. The cooling element construction is shown in Fig. 9. The investigator selected three clay layer thicknesses, *viz.*, Thickness 1–2 mm, Thickness 2–4 mm and Thickness 3–6 mm. Thickness 1 was found to be the most operative in transferring the heating load of a wooden enclosure that contained the copper specimen in natural convection and hence selected this much thick clay to place on the PV module back. The PV module with clay showed a maximum of 19.4% improvement in voltage output with an increase of 19.1% in power output as compared to a bare PV module.

The idea of bionic water transport mechanism of the trees has been perceptively used by Drabiniok and Neyer (2014) in cooling PV cells using microporous evaporation foils. Bionic cooling system emulates the water transport mechanism of trees and the mechanism of human perspiration. The researchers used a thin microstructured polymer foil and allowed water to evaporate through its micro pores on the back side of the PV modules. The design of the system is portrayed in Fig. 10.

The cooling performance of this system under calm air and heated environment, as claimed by the authors, is equivalent to

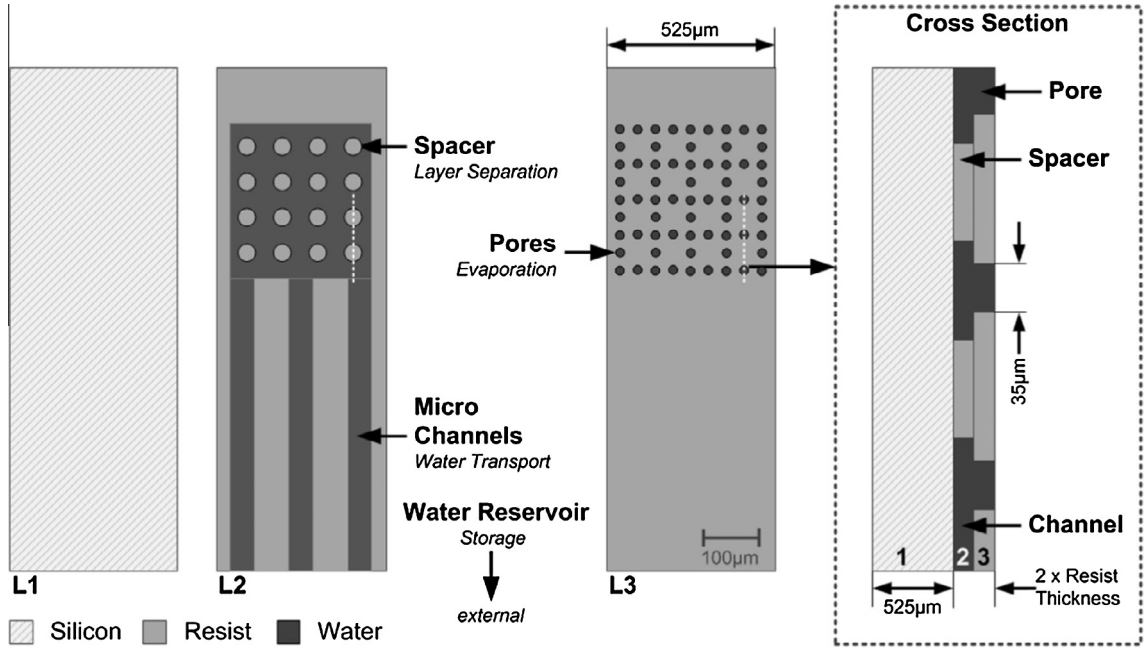


Fig. 10. Design for layer composition of the microporous evaporation foil (Drabiniok and Neyer, 2014).

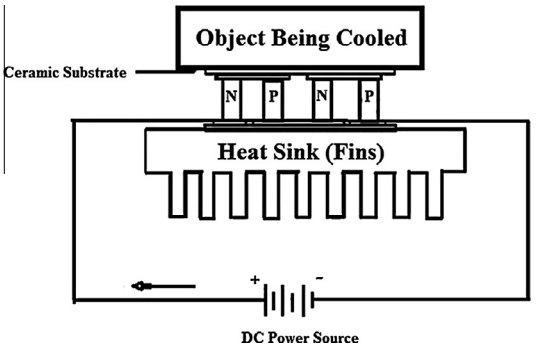


Fig. 11. Schematic of the thermoelectric cooling system (Najafi and Woodbury, 2013).

the other conventional cooling methods for PV modules. A reduction in temperature up to 11.7 °C was claimed by the authors. Moreover, once the cooling process starts, the system is self-driven by solar energy and do not need any extra energy input.

A very distinct cooling method for PV cells proposed by Najafi and Woodbury (2013) used a thermoelectric cooling (TEC) technique based on the principle of Peltier effect. A TEC module which is run by a part of the power produced by the PV cell has been joined at the rear of the PV cell. The schematic of the thermoelectric cooling system is shown in Fig. 11.

The authors simulated the thermal and electrical behavior in MATLAB and ascertained the values of temperatures in various portions of the system. Hence, authors calculated the required power required for the functioning of the TEC module and the additional power generated by the PV cell by cooling. The authors claimed that a minimum module temperature of 344.41 K (when the ambient temperature was 350.45 K) was obtained with an extra power output of 0.0704 W. The system is believed to be more operative under high solar radiation levels.

4.2. Passive cooling with phase change materials

Phase changing materials (PCM) are substances that have a high heat of fusion, melt and solidify at a narrow temperature range and are capable of storing and releasing huge quantities of heat. Energy efficient building envelopes are already made with PCM and now it has an immense potential to apply for PV cooling.

A three-dimensional numerical model has been devised by Huang et al. (2007) to study the influence of PCM in PV for limiting the temperature rise. The fluid flow adjacent to the side end faces decelerates due to the non-slip boundary condition. Therefore, the effect of heat transfer from the side faces becomes substantial. Also, the possibility of the use of pin fins into the bulk of PCM has been reported in this work. It was found that pin fins improve heat transfer into PCM and enhance thermal consistency but create an obstacle to natural convection.

The yearly energy yield attained by a PV system integrated with PCM layer has been examined by Smith et al. (2014). The PCM block that acts as heat sink restricts the limit of the peak temperature of the PV module thus up surging the efficiency. Energy fluxes into and out of the PV-PCM system is shown in Fig. 12. The authors showed that the improvement in energy output with

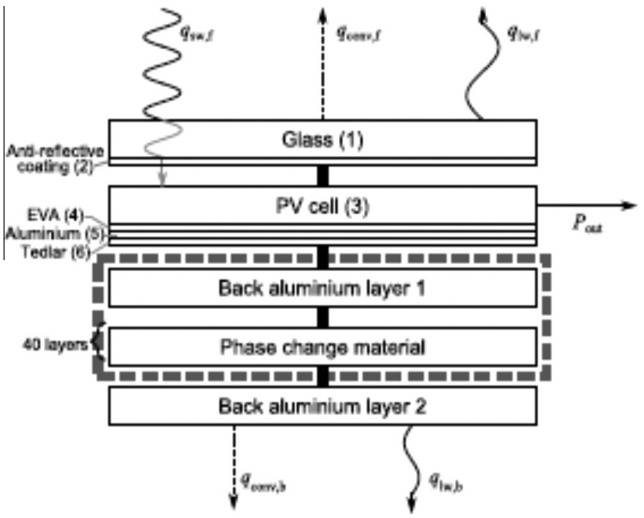


Fig. 12. The energy fluxes into and out of the PV-PCM system (Smith et al., 2014).

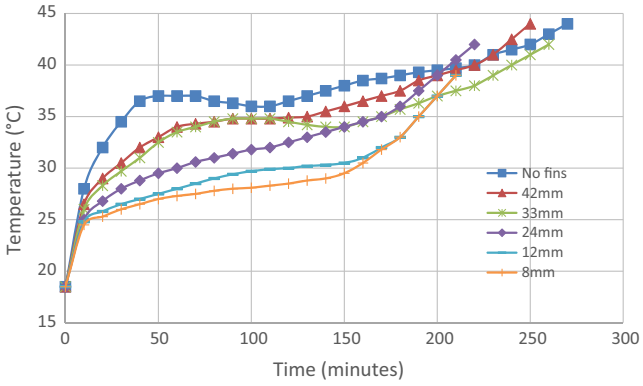


Fig. 13. Average temperature on the front of the system with various fin spacing and no fin (Huang et al., 2011).

a PV-PCM system compared to a non-PCM system is positive everywhere in the world and in some regions it yields 6% excess energy on an annualized basis.

The temperature control and heat energy storage possibility of the phase changing materials were studied by Hasan et al. (2014) under high-temperature conditions. A drop in temperature of 12 °C was achieved. The voltage gain against this temperature drop was 0.4 V as observed in the study. Results also show that such systems are effective in mild weather conditions where the PCM always return to its solid phase. An internally finned heat sink charged with phase change material for thermal regulation of PV has been designed by Huang et al. (2011). They studied the effect of natural convection and crystalline segregation of PCM and studied the thermal performance for internal fin arrangements. The porosity formed in the central bulk PCM when solidified at –25 °C. For the same liquid volume, PCM RT35 has less solid volume than PCM Waksol A, consequently, the cavity formed in the RT35 sample (sky blue color) was larger than that in the Waksol A sample (orange color). PCM passively bound the PV temperature to rise, but small heat transfer rates and petty heat removal performance of PCM in a molten state is the main challenge. The authors studied the thermal performance of PV-PCM systems with and without fin arrangements. Without fin a PV temperature of 42 °C was obtained in 250 min.

Through the use, the phase changing material RT 27 with interior fins, the temperature rise of the PV is lower in comparison with a single flat aluminum plate. Fig. 13 shows the average temperature on the front of the system with various fin spacing and no fins. It is revealed that the fins successfully restrict the temperature rise. Fins reduce the temperature rise as well as thermal stratification within the system. But, a negative point is noticed that fins lower the time over which temperature control remains. The load of the metal fins is also a difficulty for practical usage because the structure of the PV panel is generally not meant to withstand such extra load.

4.3. Conventional passive cooling with air

Temperature control for PV modules integrated on the rooftop with an open channel fitted underneath the module has been modeled and studied by Mittelman et al. (2009). The modules are cooled by radiation and free convection. The authors also performed the numerical simulation for the heat transfer in the system using FLUENT 6.2 software. A contribution of this work is the development of a comprehensive correlation for the average channel Nusselt number for the combined convective-radiative cooling. The authors made extrapolations of the global channel Nusselt number $\overline{Nu}^*$ numerically as a function of modified channel Rayleigh number $Ra''\sin\phi$ for various aspect ratios and for both pure convection and combined convection-radiation. A correlation for global Nusselt number has been suggested by the authors from the numerical data as below (Mittelman et al., 2009):

$$\overline{Nu}^* = F(L/S).(Ra''\ \sin\phi)^{0.203} \tag{27}$$

For $\varepsilon_1 = \varepsilon_2 = 0.9$, $Ra'' \leqslant 10^8$. $15 \leqslant L/S \leqslant 50$ and $30° \leqslant \phi \leqslant 90°$ where the function F depends on the aspect ratio L/S and is given by

$$F(L/S) = -3.38 \times 10^{-6}(L/S)^3 + 0.000687(L/S)^2 - 0.04419(L/S) + 1.833 \tag{28}$$

The authors claim that inclusion of the channel at the back of the PV panels drop the PV temperature by up to 20 K, with a subsequent gain in absolute efficiency of 1–2%, depending on the channel geometry and the solar insolation.

Sandberg and Moshfegh (2002) studied the impact of the geometry of the air gap and location of solar cell module in the performance of buoyancy-driven air stream in PV facades. The authors analytically derived the relations for mass flow rate, velocity, temperature and location of neutral height in air gaps behind solar cells placed on vertical facades. The authors reported that for turbulent flow and constrained flow (i.e., flow affected by losses at inlet and outlet), the mass flow rate, velocity and volumetric flow rate follow a power-law relation with the actual heat input raised to an exponent of 1/3 whereas the temperature boost between inlet and the outlet is proportional to the heat input raised to 2/3.

4.4. Passive liquid cooling

A passive thermal management system that employs heat spreaders in combination with cotton wick has been devised by Chandrasekar and Senthilkumar (2015). The setup is shown in Fig. 14. The maximum temperature of the PV module was lowered from 49.2 °C to 43.3 °C which is about 12%. This is achieved by the combined effect of evaporative cooling and the fin effect of the heat spreaders at the backside of the PV module. With the lessening in PV temperature, the electrical output was improved by 14%.

The possibility of passive cooling by attaching wet cotton wick coil at the rear of the standalone PV module was investigated by Chandrasekar et al. (2013). The scheme is illustrated in Fig. 15. The efficiency pattern of the PV module under different operating conditions is presented in Fig. 16 along with the radiation data. A maximum reduction in module temperature of 20 °C (from 65 to 45 °C) was achieved which is equivalent to 30% reduction. A maximum of 10.4% efficiency has been obtained by using wick coil soaked with water. On the other hand, efficiency drops to its least of 9.0% without cooling. In the case of cooling with wick structure seeped in Al_2O_3/water and CuO/water nanofluid, the PV module efficiency obtained was 9.8 and 9.4% respectively. The maximum rise in electrical efficiency is obtained with the use of water in numeral which is 15.5%.

The important outcomes of researchers on passive cooling systems have been summarized in Table 2. It may notice from the Table that Chandrasekar et al. (2013) achieved the highest reduction in temperature of 20 °C with passive cooling by cotton wick structure.

5. Active cooling technologies for solar PV systems

Active cooling is a feasible way to cool PV modules under certain modes of application. The major benefit with active cooling technologies is coming from its thermal energy harvest, not from the electrical output.

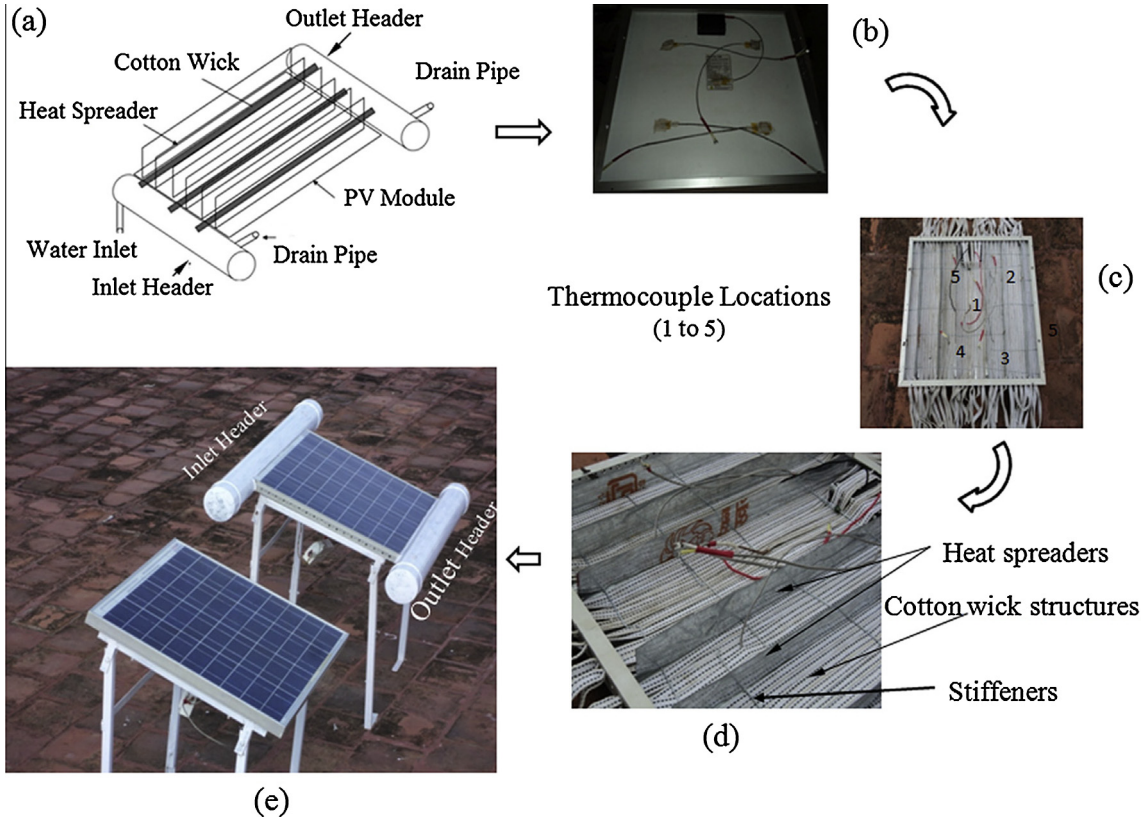


Fig. 14. (a) Heat spreader and cotton wick at PV module back (schematic), (b) thermocouple positions, (c) heat spreader and cotton wick at PV module back (photograph), (d) cotton wick attachment to heat spreaders by stiffeners, (e) experimental set up with and without cooling arrangement (Chandrasekar and Senthilkumar, 2015).

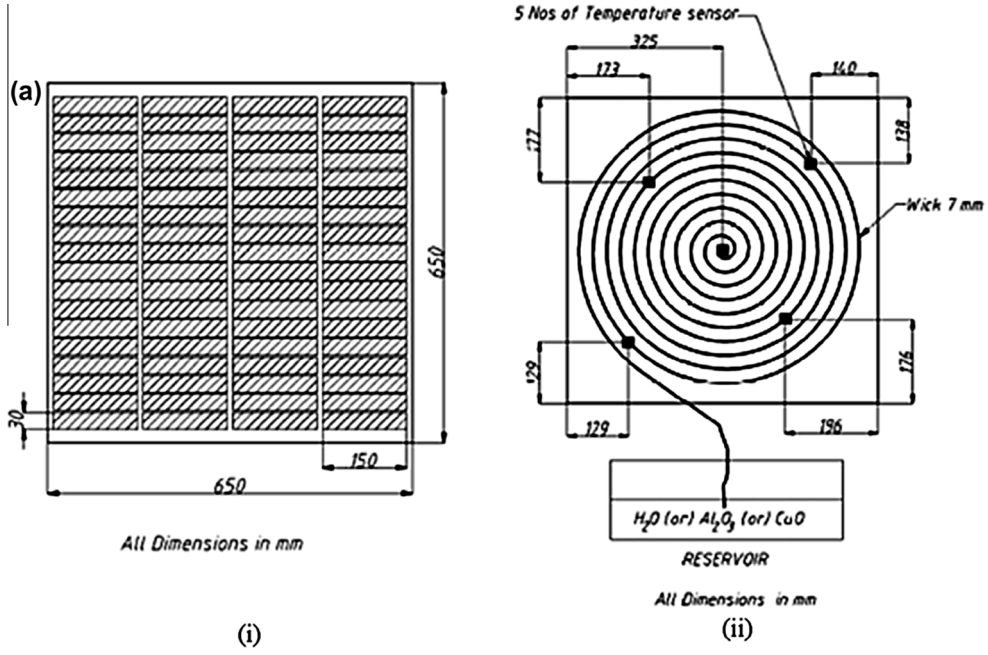


Fig. 15. (i) PV module dimensions, (ii) rear side with wick structure (Chandrasekar et al., 2013).

5.1. Liquid active cooling

Liquids such as water have a high heat carrying capacity and high thermal conductivity compared to air, thereby results in a high value of heat transfer (Prakash, 1994). But the system using liquid coolant needs to be air-tight and corrosion resistant. Water is applied as coolant liquid in most cases due to its wide availability and other favorable properties. Although PV systems integrated with water cooling facility are costlier than the air cooled systems, in industrial production sectors water cooled systems are more

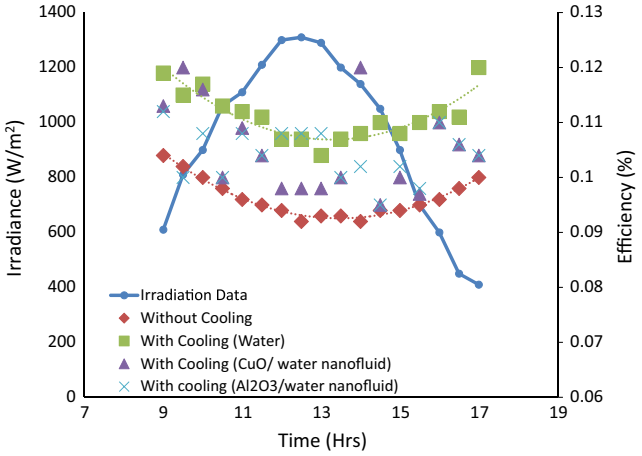


Fig. 16. Efficiency pattern of the PV module in diverse operating conditions (Chandrasekar et al., 2013).

suited as the heat extracted can be used as process heat thereby curbing some expenses.

A cooling technique that employs simultaneous water supply on both sides of the PV panel has been reported by Nižetić et al. (2016). The increase in PV panel total electric power output and total electrical efficiency found to be 16.3% and 14.1% respectively under peak solar irradiation. In addition, the panel temperature was decreased from an average 54 °C (non-cooled PV panel) to 24 °C. The principal achievements of this cooling technique are shown by the following Figs. 17 and 18. From Fig. 17, it is clear that when both sides of the panel are cooled at the same time, efficiency improves the most and panel mean temperature falls to its lowest. The reduction in temperature achieved between non-cooled PV and cooled PV panel is shown in Fig. 18. The panel temperature ranged from 52 to 60 °C and mean temperature was around 54 °C.

The authors presented a brief summary of their notable outcomes depending on different cooling regimes in Table 3 as follows. It can be observed from the table that simultaneous cooling of both back and the front side of the PV panel yield the maximum effective increase in electrical efficiency of 5.9% while lowers the average panel temperature as low as 24.1 °C. So, this simultaneous cooling technique may be a good solution for PV thermal management.

An innovative approach of using converging channel heat exchanger for PV cooling has been presented by Baloch et al. (2015). The researchers performed an experimental and numerical investigation to acquire a low and even temperature on the surface of PV panel under extreme environmental conditions of Saudi Arabia for the month of June and December. The schematic of the heat exchanger is shown in Fig. 19. The CFD analyses are done for seven different converging angles from 0 to 10°. A converging angle of 2° found to be the best in terms of temperature distribution and average cell temperature. Significant reduction in cell temperature from 71.2 °C (non-cooled) to 45.1 °C (cooled with converging channel) in June and that from 48.3 to 36.4 °C for December was observed. Improvement in power output was found to be 35.5% maximum while the conversion efficiency was upgraded by 36.1% as compared to the non-cooled PV system. In addition to thermal effect, the authors also performed economic analysis using Levelized Cost of Energy (LCE) for the laboratory scale set up for a non-cooled and cooled PV wherein the relative Levelized cost of energy was found to reduce by 19.5%.

Table 2
Summary of the review of the passive cooling systems.

Researchers [Ref.]	Technology adopted	Performance upsurge	Temperature aspects	
			PV module temperature	Reduction in temperature
Zhu et al. (2014)	Radiative cooling by using very thin patterned silica glass			13 °C (in PV module temperature)
Ebrahimi et al. (2015)	Cooling by natural vaporization	7.3% increase in power output	48.3 °C (with low flow rates of vapor) 39.3 °C (with high flow rates of vapor)	
Chandrasekar and Senthilkumar (2015)	Heat spreader in conjunction with cotton wick	14% increase in electrical yield	43.3 °C	5.9 °C (in PV module temperature)
Alami (2014)	Evaporative cooling using synthetic clay covered copper sheet	19.4% enhancement in voltage 19.1% enhancement in power output		15 °C (in PV enclosure temperature)
Drabiniok and Neyer (2014)	Evaporation using bionic microporous foils	Equivalent to conventional methods		11.7 °C (in PV enclosure temperature)
Chandrasekar et al. (2013)	Diffusion of fluid by cotton wicks on PV panel backside	15.5% increase in electrical efficiency (with water)	45 °C	20 °C (in PV module temperature)
Najafi and Woodbury (2013)	Thermoelectric cooling (TEC)	0.0704 W extra power output	344.41 K (when ambient temperature is 350.45 K)	
Smith et al. (2014)	Latent heat storage by phase change material	Maximum 6% enhancement in power output		
Hasan et al. (2014)	Latent heat storage by phase change material	0.4 V gain in voltage		12 °C (in PV module temperature)
Huang (2011)	Latent heat storage by phase change material		42 °C (in 250 min, without fin)	
Huang et al. (2007)	Latent heat storage by phase change material			
Ryan and Burek (2010)	Buoyancy-driven air flow in solar chimney			
Mittelman et al. (2009)	Radiation and free convection	1–2% gain in absolute efficiency		10–20 K (in PV module temperature)
Sandberg and Moshfegh (2002)	Buoyancy-driven air flow in PV facades			

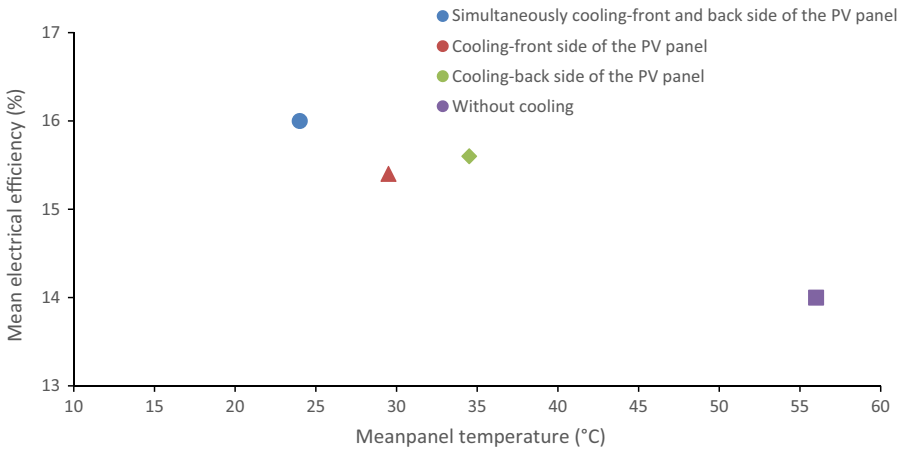


Fig. 17. Mean maximal PV panel efficiency as a function of mean panel temperature for different cooling options (Nižetić et al., 2016).

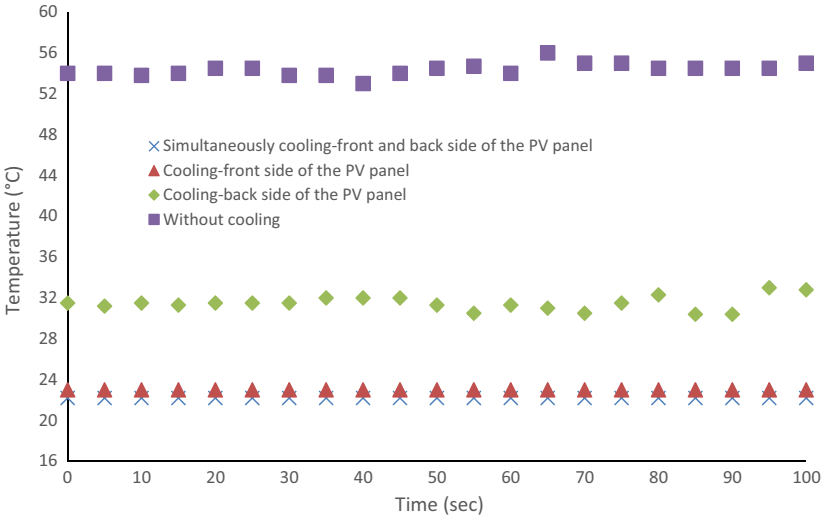


Fig. 18. PV panel temperature reduction for different cooling regimes under highest solar irradiation (Nižetić et al., 2016).

Table 3
PV panel means performance parameters for different examined cooling circumstances (Nižetić et al., 2016).

Applied cooling option	Maximal power output (W)	Relative increase in power output (%)	Effective increase in power output (%)	Average panel temperature (°C)	Electrical efficiency (%)	Effective increase in electrical efficiency (%)
Without cooling	35	–	–	56	13.92	–
Back surface cooling	39.9	14.0	5.4↑	33.7	15.59	3.6↑
Front surface cooling	40.1	14.6	6.0↑	29.6	15.42	2.5↑
Simultaneous cooling	40.7	16.3	7.7↑	24.1	15.92	5.9↑

The application of micro-channels in water cooling systems is gaining popularity. Rahimi et al. (2015) employed two types of micro-channels, single-header and multi-header with the equal hydraulic diameter to convey water for PV cooling. In Fig. 20, it is evident that there is a notable reduction in average PV cell temperature at the lower range of the water flow rate. Moreover, it is revealed from this Figure that the average temperature of the PV cell for single-header micro-channels was higher than the multi-header one. The results showed that heat removal is 19% greater and generated power is about 28% higher in multi-header channel as compared to the single-header one. It was also established that the minimum PV cell surface temperature for multi-header microchannel was 44 °C as compared to that for single header microchannel of 47 °C.

Water immersion cooling technique for PV cooling under extreme temperatures was experimentally demonstrated by Mehrotra et al. (2014). Solar cell dipped in water was examined under actual climate. In Table 4 is given a comparative performance depiction of the panel immersed at various depths under water.

The authors reported that panel efficiency had been improved by 17.8% at a water depth of 1.0 cm. They also suggested that this

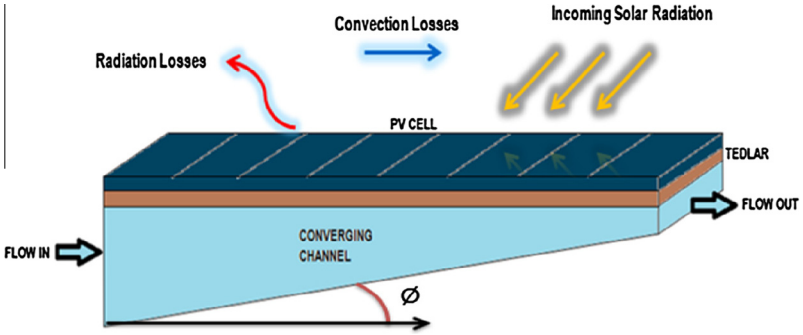


Fig. 19. Schematic of the converging channel heat exchanger with heat transfer modes (Baloch et al., 2015).

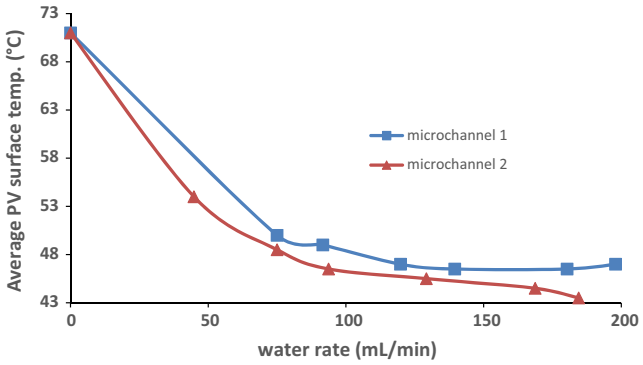


Fig. 20. Comparison of PV surface temperature profile using two types of microchannels against water flow rate (Rahimi et al., 2015).

Table 4
Performance of the PV panel at different depths under water (Mehrotra et al., 2014).

Depth (cm)	Average surface temperature (°C)	Average output voltage (V)	Average electrical efficiency (%)
0	59.4	8.011	4.04
1	34.8	8.75	4.76
2	32.4	8.79	4.44
3	33	8.8	4.311
4	31.2	8.811	4.211
5	31.8	8.8	4.16
6	30.8	8.78	4.03

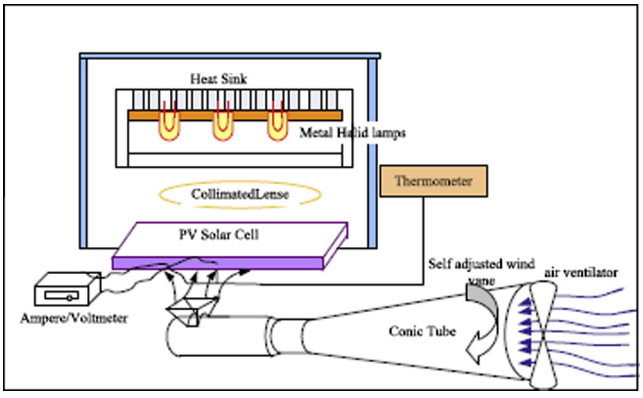


Fig. 21. Schematic diagram of the self-adjusted jet impingement system for PV cooling (Rahimi et al., 2014).

technique can also be applied to larger water reservoirs like rivers, lakes, canals, even oceans few centimeters underwater and set on big floating frames. As the availability of wide and even land area is becoming scarce day by day so this technique may be a good solution with minimal environmental impact.

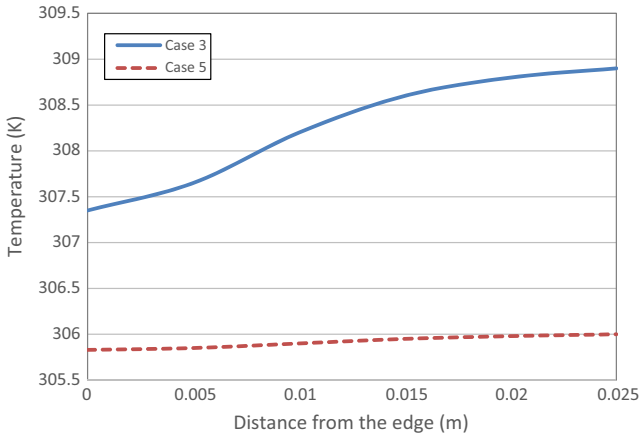


Fig. 22. Temperature profile of PV cell for PV-water cooling system with aluminum ducts with (Case 5) and without (Case 3) metal sheet (Arcuri et al., 2014).

A self-adjusted jet impingement system to cool PV module was designed and developed by Rahimi et al. (2014). In fact, the authors introduced a new technique to join up the wind and photovoltaic cell to generate electrical energy from both the wind and solar cell simultaneously. The scheme proposed by the investigators has been illustrated in Fig. 21. Although the average temperature of the combined system is slightly high (only 3 °C) than the simple cooling system without the turbine, the total power generated from PV cell and wind turbine was found to be 21% more than that with the simple cooling system. The cell PV cell temperature was obtained 57.6 and 85.4 °C under solar intensities of 570 and 910 W/m^2, respectively. The PV temperature reduction varies from 15 to 26 °C, depending upon the solar intensity applied.

Arcuri et al. (2014) examined the effect of both air and water cooling systems on the electrical efficiency of a PV panel. Firstly, the authors analyzed the usual PV panel with the help of finite element software TRNSYS and then carried out experimental investigations using both water and air. They found that for the water system, cooling performance is extremely influenced by pipe material; the overall heat transfer coefficient for metal pipes is always greater than 40 W/m^2 K. Moreover, they observed that the turbulent flow condition compared with laminar flow take along significant favorable effects especially for the ducts formed from metal. The use of aluminum sheet along with the metal pipes boosts the heat rejection capability of the PV panel onto the aluminum frame using its high thermal diffusivity. Fig. 22 shows the temperature profile of the PV cell for PV-water cooling system with aluminum ducts with and without a metal sheet.

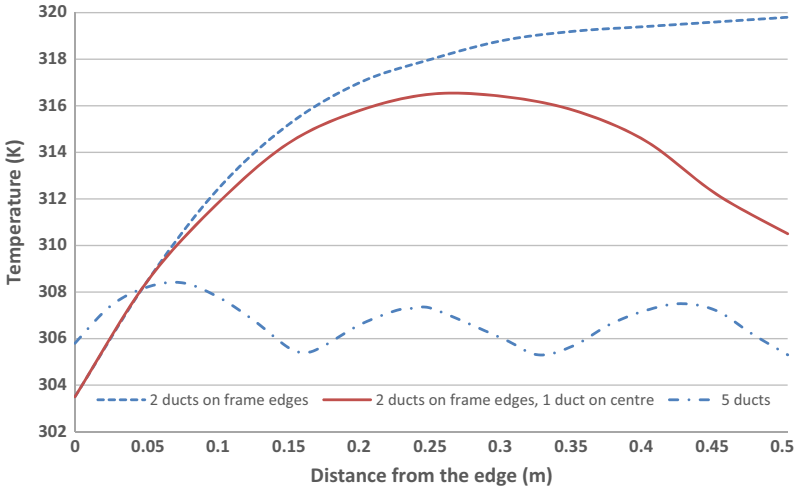


Fig. 23. Temperature profile of PV cells for various PV-water cooling system regarding the optimization phase (Arcuri et al., 2014).

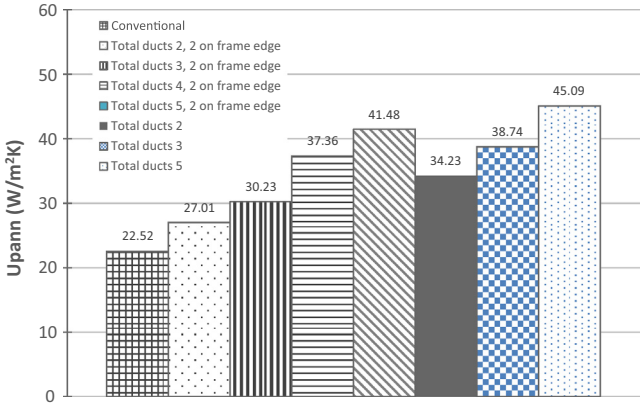


Fig. 24. Overall heat losses coefficient of PV-cooling water system for the solution concerning optimization phases (Arcuri et al., 2014).

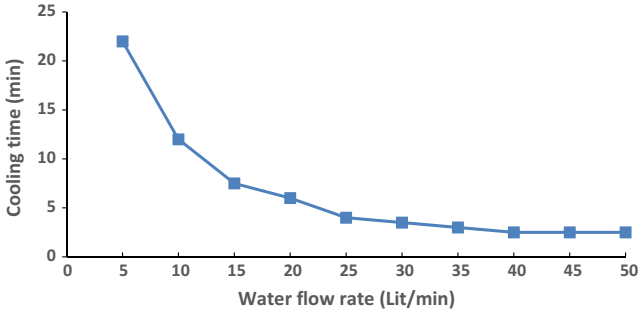


Fig. 25. Cooling time versus water flow rate (Moharram et al., 2013).

The temperature characteristics of some PV-water cooling system configurations concerning the optimization phase are shown in Fig. 23. Three configurations were tested by the author: two ducts located at the PV frame edges, two ducts located at the PV frame edges and one at the center and five ducts placed equidistant from the center. For the first configuration with two ducts on the edges of PV frame, the temperature at the middle of the panel reaches close to NOCT value. Adding an extra duct at the middle of the PV panel decrease the cell temperature and the highest value remains below the NOCT.

The cell temperature profile of the optimized configuration, in the case of five ducts, placed equidistant from the center, has a maximum variation of 3 K. Temperature is somewhat higher near the frame due to the nonexistence of cooling ducts in this area. The overall heat losses coefficient of the whole systems as a function of ducts number and ducts location is shown in Fig. 24. The ducts in each solution are equidistant from the center of the PV panel. It can be seen that heat loss coefficient is maximum (45.09 W/m^2 K) for a total of five ducts.

A dynamic water cooling augmented with the sun-tracking system under elevated temperature was carried out by Rodgers and Eveloy (2013). Four similar monocrystalline 32-cell PV modules with a rated peak power output of 140 W were electrically and thermally analyzed to evaluate the impact of module cooling and module orientation. An experimental study was carried out under steady-state and dynamic cooling situations to compare the influence of continuous and intermittent water-cooling. The results show the effectiveness of water-cooling along with sun tracking to augment PV module power output in the prevailing climatic conditions. Steady-state water cooling of a stationary module with un-chilled water (35–40 °C) proved less effective with an improvement in electrical power output by 23%. But, use of chilled water (7–20 °C) improved the output up to 40%.

Water cooling of PV systems under extreme environments has been addressed in solar PV research. Bahaidarah et al. (2013) performed numerical analysis and experimental study of an isolated PV and PV-water cooled hybrid system regarding its electrical and thermal performance in climatic conditions of the desert. The authors have found that energy compilation with the hybrid PVT water collector system is higher (about 4 times) than that with an isolated PV system. At an irradiance level of 900 W/m^2, the hybrid PVT system arrested 750 W whereas the PV only system harvested nearly 190 W. With active cooling by water, the functional temperature of the module drops notably from 45 to 34 °C which is a drop of 20% and the electrical efficiency increases of 9%.

Moharram et al. (2013) have done an investigation, the main objective of which was to minimize the use of water and hence to optimize the electricity consumption in PV cooling. This method can be efficiently applied in the hot arid regions or desert areas. It was noticed that the temperature coefficient at the maximum power operating point is – 0.5%/°C and without cooling, a rise in 10 °C caused waning efficiency of the cells by 12.5%. On the other hand, as the cooling system was operated for 5 min a decline in the solar cell temperature by 10 °C and raise in efficiency by 12.5% was almost recovered. In addition, the panel was cleaned at same time of cooling operation with the water which improves

the efficiency more. This facility is very much significant for hot and sandy regions where sand storms can shroud the panel with a film of dust and consequently dim the solar radiation and depreciate the efficiency of the panels. The most interesting verdict of this research is the selection of maximum allowable temperature (MAT) which facilitates to operate the panels at an optimum efficiency level with the least amount of water usage. The authors determined the optimum MAT is for desert climate as 45 °C. The optimum cooling water flow rate has also been depicted as shown in Fig. 25. The high flow rate cools the panel over a shorter period compared to lower flow rates, but the cost for the water pump operation increase at the same time. The cooling time is reduced as the water flow rate is increased. The optimum flow rate estimated was from 15 to 25 l/min. The further increase of flow rate does not give remarkable benefit to the cooling effect.

Odeh and Behnia (2009) performed a long-standing performance modeling of solar water pumping systems. Cooling was accomplished by dripping water on the top surface of the panel. The PV module surface temperature in a representative summer day was found to reach approximately 58 °C and the rise in cell temperature above a standard operating temperature was observed to be 45 °C which caused a drop of 5% in output power. Water cooling decreased the cell operating temperature around 26 °C under a radiation level of 1000 W/m^{2}and an excess in the power of about 15% was achieved. The authors also suggested that cooling by underground water in mid of the day is more efficient than cooling with stored water. The economic feasibility study revealed that the cost of extra cooling configuration was 1.7% of the first cost of the water pumping system and the payback period was estimated to be 2.5 years.

5.2. Air active cooling

Air cooling offers less expensive cooling of PV systems where heat can be removed by either natural convection or forced convection of air. Air is the preferred media for low-cost cooling despite its poor thermo-physical characteristics (Tonui and Tripanagnostopoulos, 2007a).

Sahay et al. (2015) reported a newly developed method named as ground coupled central panel cooling system (GC-CPCS). The projected scheme serve to cool the solar panels by forced convection of air driven by a blower, the blower being run by another dedicated PV panel. Air flows through a ground-coupled heat exchanger and decreases its temperature. The cooled air soothes the solar panels while passing beneath them. The researchers installed nine solar panels of 100 W each. The air is flown by a single blower and the cold air is distributed to each solar panel through the pipe. Nozzles are attached to the pipes in order to ensure that streamline flow in desired directions. The author reported marked improvement in conversion efficiency using GC-CPCS.

Kaiser et al. (2014) performed an experimental investigation on the effect of the air gap size and the forced convection provoked by the building ventilation system on the cell temperature and accordingly on the electric efficiency of the PV panel in BIPV, for diverse values of irradiation and various atmospheric temperatures. From the experimental investigation, the authors found that the solar irradiance and the prompted velocity have got significant roles on the temperature of the PV module. The authors proposed some important correlations such as correlation for Ross coefficient, k, cell operating temperature, T_c, electrical efficiency, η_c and power output, P which are validated by previously established results. Some of the correlations of interest are as follows:

Ross coefficient (Kaiser et al., 2014),

$$k = 0.045(1 + V_v)^{(-6.311b/L - 0.162)} \tag{29}$$

where V_v is the induced velocity and b/L is the aspect ratio of the air channel.

Cell operating temperature (Kaiser et al., 2014),

$$T_c = T_a + 0.045(1 + V_v)^{(-6.311b/L - 0.162)} G_T \tag{30}$$

where T_a is the ambient temperature and G_T is the solar irradiance.

Electrical efficiency (Kaiser et al., 2014),

$$\eta_c = \eta_{Tref}\left[1 - 0.006\left(T_a + 0.045(1 + V_v)^{(-6.311b/L - 0.162)} G_T - 25\right)\right] \tag{31}$$

where η_{Tref} is normally specified by the manufacturer (otherwise take as 0.15), β_{ref} and T_{ref} are taken as 0.006 and 25 °C.

Electrical power output (Kaiser et al., 2014),

$$P = \eta_{Tref} A_s G_T\left[1 - 0.006\left(T_a + 0.045(1 + V_v)^{(-6.311b/L - 0.162)} G_T - 25\right)\right] \tag{32}$$

where A_s is the aperture surface area of the module?

The authors presented the impact of induced velocity on the power output for a definite ambient temperature of 25 °C and an aspect ratio of 0.0825, as shown in Fig. 26. It can be seen that as the induced velocity is amplified from natural ventilation to a magnitude of 6 m/s the power output is increased by 19.13%.

Experimental performance study of a PV thermal air collector has been conducted by Kim et al. (2014). Forced ventilation of air

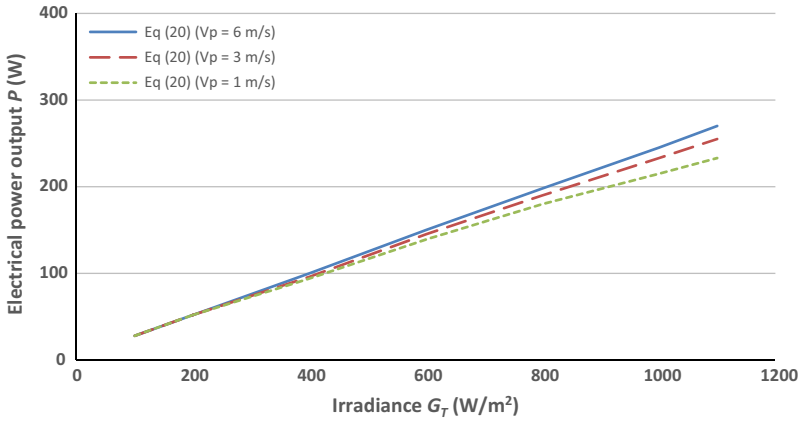


Fig. 26. Effect of the induced velocity on the electrical power at diverse irradiances (T_a = 25 °C, b/L = 0.0825) (Kaiser et al., 2014).

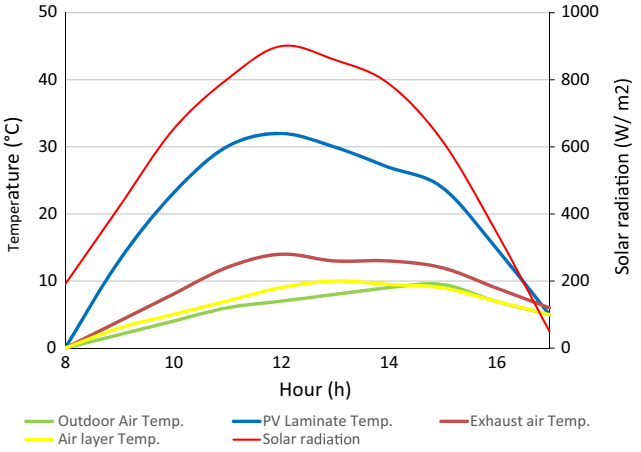


Fig. 27. Temperature of PV module, air layer and exhaust air (Kim et al. 2014).

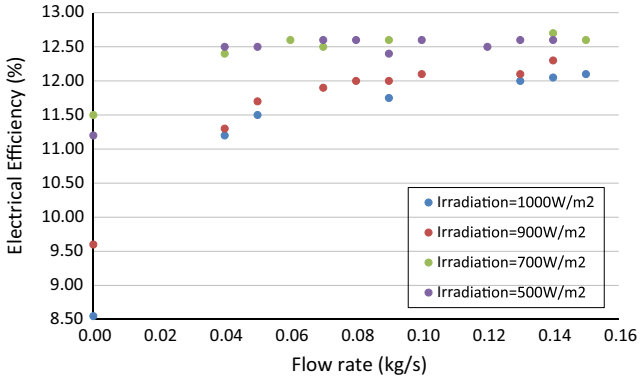


Fig. 29. Effect of flow rate on electrical efficiency (Teo et al., 2012).

at a rate of 240 m^3/h was maintained by means of a fan installed in the exhaust air pipe. It is evident from Fig. 27 that temperature of the PV module could be retained low at 12–32 °C because of the heat carried away by the exhaust air which reached 3.5–14 °C while the ambient temperature was – 1.6–9.5 °C. So, the heated air from the air collector had around 5 °C elevated temperature than the ambient air. The authors reported that the thermal and electrical efficiency of the system were about 22 and 15%, respectively. The electrical efficiency of the PV panel at STC is 15.46%, which indicates that the performance of the hybrid photovoltaic thermal (PVT) air collector is very near to the PV the only system.

Mazón-Hernández et al. (2013) examined the effects of free and forced convection of air through various geometries of ducts on the electrical parameters of a PV module. For this purpose, two photovoltaic panels were used, one panel (panel A) is isolated to be used as a reference and the other (panel B) with an air channel beneath the panel varying in space. It was found that in the case of natural convection, the modified panel is warmer than the isolated one, the temperature difference between panels being higher for the smallest air channel. But in the case of forced convection, the electrical yield is higher in the modified panel than the isolated one due to the boost in the heat transferred to the air flow by forced convection. The effect of channel aspect ratio on electrical parameters were also investigated (Fig. 28). When the aspect ratio is higher the open circuit voltage increases, whereas the short-circuit

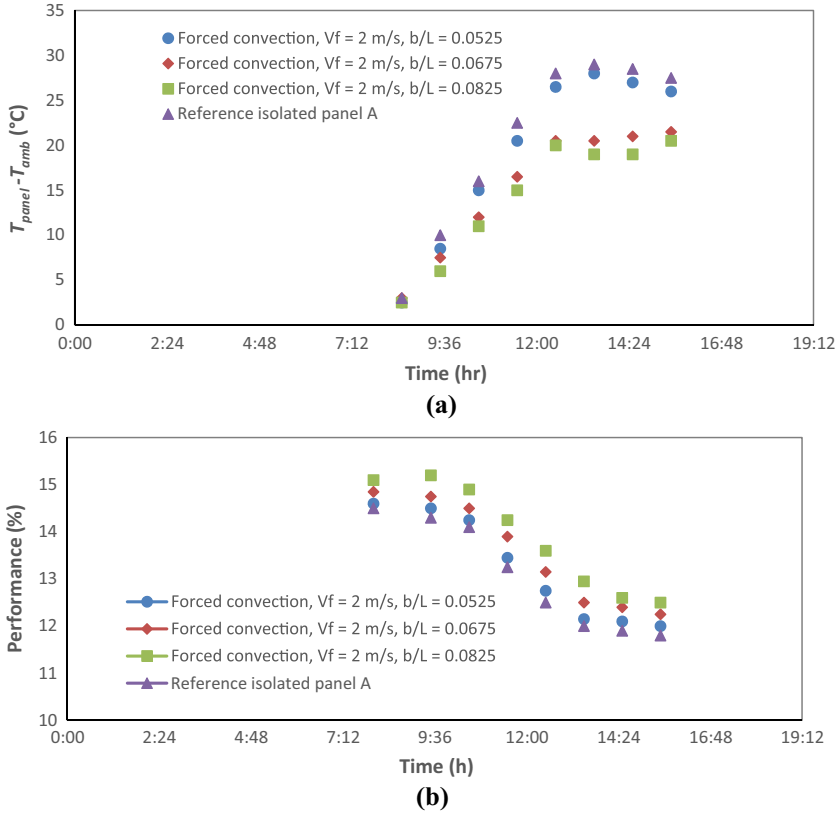


Fig. 28. Effect of aspect ratio and forced velocity on the panel temperature and its performance throughout the day (Mazón-Hernández et al., 2013).

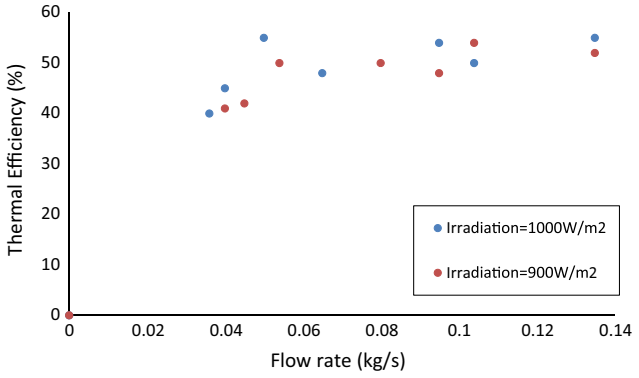


Fig. 30. Effect of flow rate thermal efficiency (Teo et al., 2012).

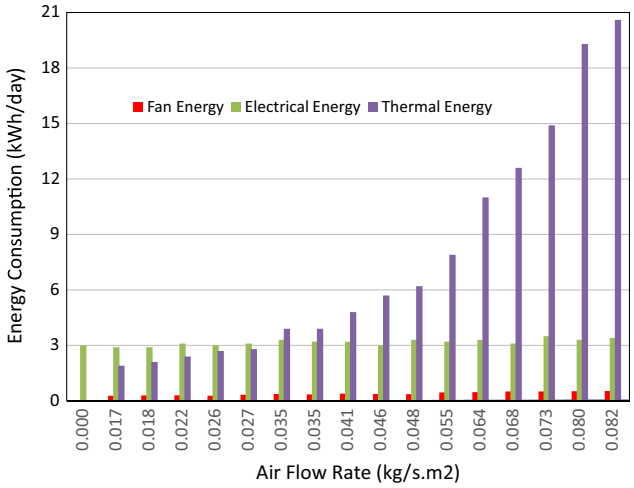


Fig. 32. Daily PVT system energy output and fan power versus air mass flow rate (adjusted for 5.75 kW/m^2/day) (Bambrook and Sproul, 2012).

current decreases and the peak power enhances up to 7.5% due to the lower temperature (refer to Fig. 28(b)). For identical aspect ratios the same elevated irradiance, the electrical output of the panel cooled by forced convection is higher than that obtained by free convection. For a given aspect ratio, the electrical yield of a PV panel cooled by forced convection is 3–5% higher than by free convection and improves with velocity inside the air duct. The PV panel temperature decreases by 10–16 °C.

Hybrid photovoltaic thermal air collector performance was characterized through simulation and experiment by Teo et al. (2012). They employed CFD software FLUENT to characterize the flow at the manifold. Moreover, a heat transfer model was developed using finite element software COMSOL Multiphysics®. The temperature of PV module, with active cooling, increase by 1.4 °C for every 100 W/m^2 rise in solar irradiation while for passive cooling this value is 1.8 °C. The influence of air flow on electrical efficiency and that on thermal efficiency are presented in Figs. 29 and 30. The authors found the electrical efficiency to be steadier than the thermal efficiency in 5-day long test although the magnitude the former one lags far behind. The average electrical efficiency is around 10.1–10.9% whereas thermal efficiency varies between a maximum of 60 to a minimum of 40%. Therefore, the total efficiency of the hybrid system is around 50–70%. The temperature profile as predicted by the heat transfer model in COMSOL Multiphysics® shows that the maximum temperature of the system is found in the cells which are about 56.2 °C.

An improved model for a regular photovoltaic thermal (PVT) air collector has been developed and tested by for two preferred cases

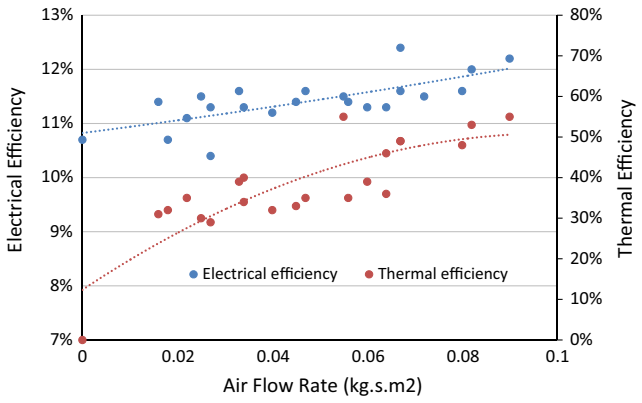


Fig. 31. Effect of air mass flow rate on the instantaneous electrical and thermal efficiency values (at 12:00 pm) (Bambrook and Sproul, 2012).

in Amori and Taqi Al-Najjar, 2012. It comprises of two similar PV modules of monocrystalline silicon connected electrically in parallel and thermally in series orientation. An insulated air duct is provided below PV modules. Fans are used for blowing the air inside the duct. The authors found better gain, collector efficiency and heat removal factors in the summer than those in the winter and they attributed this to lower heat loss coefficient. They also reported that the heat loss is chiefly due to top losses from the collector, about 70% which is due to radiation from glass to ambient due to low wind speed. The electrical power in summer is found to be higher than that is the winter while the reverse is true for electrical efficiency. This is, according to the authors, due to the negative temperature coefficient of efficiency. The fill factor is found to remain higher in the winter than in the summer. The relative electrical-to-thermal equivalent power and efficiency are found to be greater in the winter than that in the summer. Thus, the overall equivalent efficiency in the winter was observed greater than that in the summer. The authors also reported that electrical efficiency varies between 11.67 and 13.39% during the day while thermal efficiency diverges from 1.24 to 19.42%. The overall efficiency ranged 37.73–54.76%.

Strategic thermal performance indicators in order to determine the photovoltaic thermal collector design suited to the climate that prevails in Sydney, Australia has been examined experimentally by Bambrook and Sproul (2012).

The effect of mass flow rate of air on the instantaneous thermal and electrical efficiencies at 12:00 pm is illustrated in Fig. 31. It can be observed from this figure that the PVT collector efficiency values are in between 28 and 55%. With the rise in the mass flow rate of air, the thermal efficiency rises as maximum as 55–60% in which electrical efficiency contribution is 10.6–12.2%. A clear increase can be noticed in thermal energy output and the fan energy requirement for increasing air flow rates from Fig. 32.

The additional electrical energy generated when the PV modules are ventilated compared with the non-ventilated system and the corresponding fan energy requirements are presented in Fig. 33 for a certain range of mass flow rates of air. The authors suggest that this figure may be used to provide an indication of the PVT system behavior. Generally, for mass flow rates of air up to 0.55 kg/s m^2, the fan energy requirement is less than the additional PV electricity generated.

Double pass photovoltaic thermal air heater with the vertical fin in the lower channel has been used by Kumar and Rosen (2011) who carried out an investigation into have a better understanding

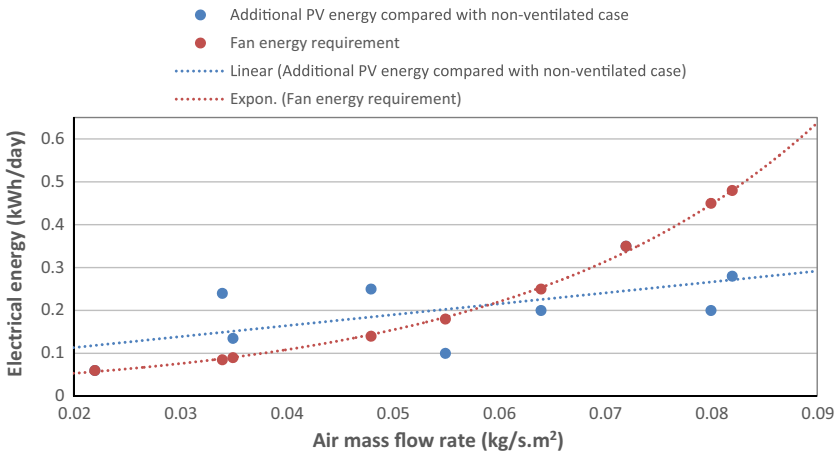


Fig. 33. Comparison of additional PV energy output and daily fan energy requirement (Bambrook and Sproul, 2012).

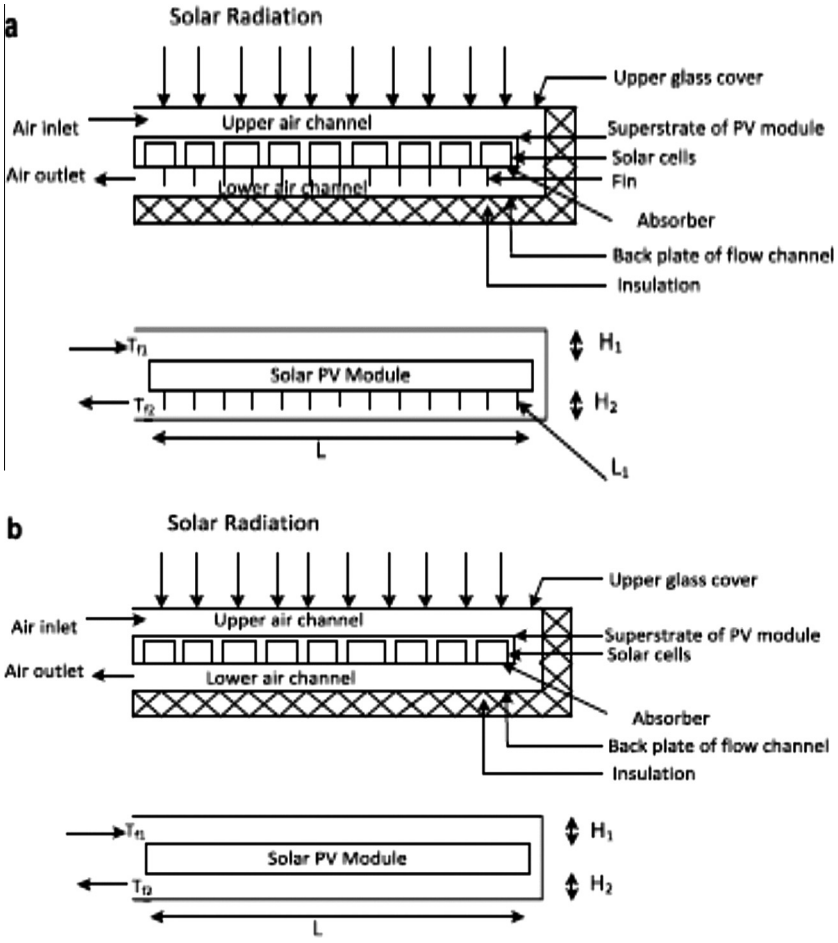


Fig. 34. Cross-sectional view of photovoltaic thermal air heater (a) with fins (b) without fins (Kumar and Rosen, 2011).

of the impact of such modifications. The air enters the upper channel of the air heater and then flows to the lower channel in the opposite direction. The cross-sectional views of the PVT solar air heater with and without fins are shown in Fig. 34.

The thermal and electrical efficiencies were improved by 15.5 and 10.5% respectively by the inclusion of fins on the rear of the absorber surface. The authors also observed the effect of packing factor on the thermal, electrical and equivalent thermal efficiencies and the rise in the air temperature. The authors claim that the extended fin area helps to lower the cell temperature noticeably from 82 to 66 °C. Moreover, the total equivalent thermal efficiency of the PVT collector increases about 17% within this range.

Sarhaddi et al. (2010b) examined the thermal and electrical performance of photovoltaic thermal (PVT) air collector through com-

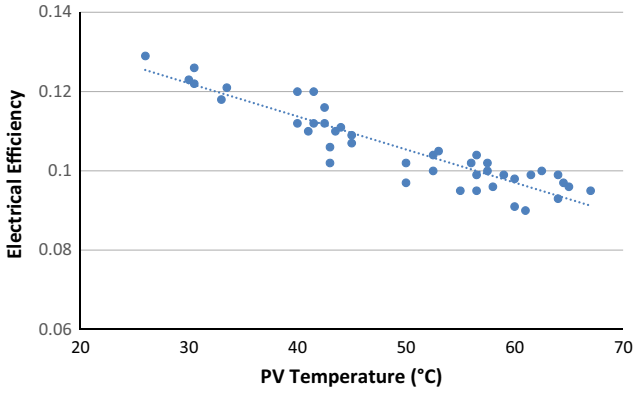


Fig. 35. Electrical efficiency as a function of PV temperature (Tonui and Tripanagnostopoulos, 2007b).

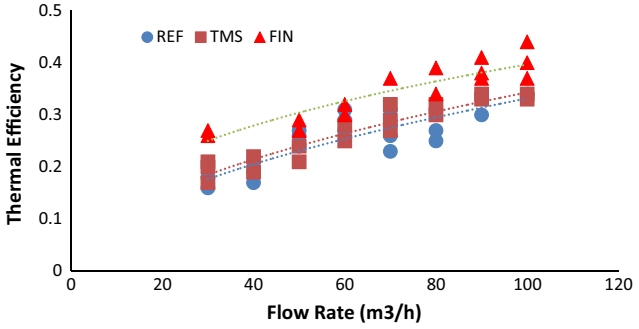


Fig. 36. Effect of air flow rate on thermal efficiency (Tonui and Tripanagnostopoulos, 2007b).

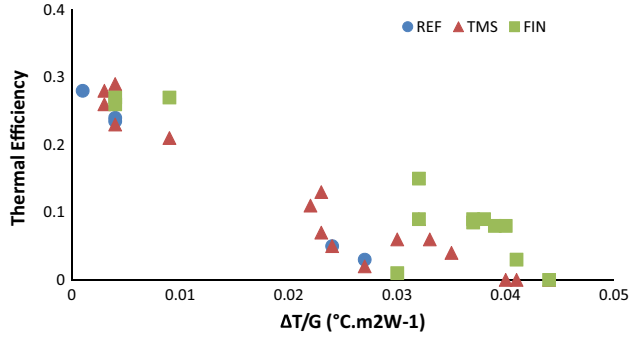


Fig. 37. Thermal efficiency as a function of $\Delta T/G$ (Tonui and Tripanagnostopoulos, 2007b).

puter simulation and parametric study. The overall efficiency of a photovoltaic thermal air collector is always superior to the thermal efficiency of a solar collector or electrical efficiency of a PV only system. As the inlet air temperature or wind speed or duct length is increased, the overall energy efficiency and thermal efficiency of a PVT air collector decrease, but when inlet air velocity increase the overall and thermal efficiency decrease. The thermal efficiency, electrical efficiency and overall energy efficiency of PVT air collector was found to be 17.18%, 10.01%, and 45%.

Photovoltaic thermal air collector with different channel depths was developed and studied by Tonui and Tripanagnostopoulos (2007b). The air flow was maintained by means of an air pump. In Fig. 35 is shown the behavior of electrical efficiency against temperature where the electrical efficiency is observed to fall with the increase in PV module temperature.

The impact of air flow rate on thermal efficiency has been predicted Fig. 36. All three systems show a similar trend of exponential increase in thermal efficiency with airflow rate where the FIN system has the highest efficiency.

The steady-state thermal efficiency gain at various air inlet temperature is plotted against $\Delta T/G$ ($\Delta T = T_{in} - T_a$ with flow or $\Delta T = T_{pv} - T_a$ under stagnation) in Fig. 37. From this figure, it can be observed that the thermal efficiency gain of FIN system is the highest with a value of 30% and in the case of REF and TMS systems, this value is 25 and 28% respectively.

The research works on the active cooling system have been summarized in Table 5. As can be seen from the table Nižetić et al. (2016) achieved the highest reduction of 30 °C in PV module temperature by simultaneous water sprinkle on both front and back side of the panel.

6. Future prospect in PV cooling

The application of the photovoltaic system is increasing in the market day by day. Although research on new photovoltaic technologies are continually in progress and efficient PV cooling technologies are expected to disembark in the market, there are still some areas which need to be addressed by the researchers. Advanced researches in following areas are eagerly awaited.

Although historically silicon (in the form of mono-, polycrystalline or amorphous) is the most used semiconductor material PV cell in the industry there have been conducted some works to find better alternatives to silicon. But still there is a wide open area to carry on thorough research work to find such materials that would not suffer voltage (thereby power) drop under elevated temperatures. Research may also be carried out to find an appropriate PV module surface coating that will facilitate more effective radiative cooling. In addition, use of phase change materials (PCM) may offer an apt solution for commercial scale passive cooling. Though several works have carried out to find an efficient way of using PCM in PV cooling, much effort is required to encourage PV-PCM technology.

Active cooling systems are very much capable ways for PV cooling even for solar farms, but the rate and amount heat dissipation is still a vital problem. Moreover, effective methods should be devised to make sure the utilization of the heat thus extracted. So, several extensive studies, both numerical and experimental, could be taken into account in order to find designs for heat exchangers that are suitable particularly for PV modules.

7. Conclusion

This paper presents an overview of recent studies of PV thermal regulation techniques. The research in progress related to PV cooling technology is summarized above, to identify the unsettled problems and the prospects in this field. Results reveal that passive systems are still feasible only for small scale usage presenting significant challenges and motivating opportunities for future research in commercial scale applications like in solar farms. PV-PCM or PVT-PCM technologies may bring a major change in passive cooling systems. On the other hand, the active cooling systems very much effective in commercial scale PV panel cooling with the shortcoming of the issue of extra power requirement. Active systems would be a legitimate solution if the heat energy harvested from the system could efficiently utilize. Search for suitable heat recovery technologies should be addressed and continually carried on to uphold this method.

Table 5
Summary of the review of the active cooling systems.

Researchers [Ref.]	Technology adopted	Performance upsurge	Cooling aspects	
			PV module temperature	Reduction in temperature
Nižetić et al. (2016)	Simultaneous water sprinkle on both sides of PV panel	16.3% (effective 7.7%) in electric power output 14.1% (effective 5.9%) in PV panel electrical efficiency	24 °C	30 °C (in PV module)
Baloch et al. (2015)	Converging channel heat exchanger	35.5% increase in power output 36.1% increase in conversion efficiency	45.1 °C (June) 36.4 °C (December)	26.1 °C (June) 11.9 °C (December) (in PV module)
Rahimi et al. (2015)	Multi-header microchannel	19% extra heat removal 28% increase in power output (as compared to single header microchannel)	44 °C	3 °C (as compared to single header microchannel)
Mehrotra et al. (2014)	Water immersion cooling	Maximum17.8% improvement in panel efficiency (at a water depth of 1.0 cm)	34.8 °C (at maximum efficiency depth)	
Rahimi et al. (2014)	Self-adjusted jet impingement (combined wind & PV module)	Power output 21% more than simple PV cooling system	57.6 °C (at 570 W/m^2) 85.4 °C (910 W/m^2)	15–26 °C (in PV cell)
Arcuri et al. (2014)	Both PVT water and PVT air collector	No specific data	No specific data	3 K (in PV cell with five cooling ducts)
Rodgers and Eveloy (2013)	Intermittent water cooling with sun tracking	40% improvement in electrical power output (with chilled water) 23% improvement in electrical power output (with unchilled water)		
Bahaidarah et al. (2013)	Hybrid PVT water collector	9% increase in electrical efficiency	34 °C	11 °C (in PV module)
Moharram et al. (2013)	Water sprinkle on front surface of the PV module	12.5% increase in efficiency		10 °C (in PV module)
Odeh and Behnia (2009)	Front surface water film cooling system	15% increase in power output		26 °C (in PV module)
Sahay et al. (2015)	Ground-coupled central panel cooling system (GC-CPCS)			
Kaiser et al. (2014)	BIPV	19.13% increase in output power (at an air ventilation rate of 6 m/s)		
Kim et al. (2014)	PVT air collector	15% increase in thermal efficiency 22% increase in electrical efficiency	12–32 °C (ambient temperature −1.6 to 9.5 °C)	
Mazón-Hernández et al. (2013)	PVT air collector with varying duct geometries	Maximum 7.5% increase in power		10–16 °C (in PV module)
Teo et al. (2012)	PVT air collector	Electrical efficiency 10.1–10.9% Maximum thermal efficiency 60% Total hybrid efficiency 50–70%	56.2 °C	
Amori and Taqi Al-Najjar (2012)	PVT air collector	Electrical efficiency 11.67–13.39% Thermal efficiency 1.24–19.24% Overall efficiency 37.73–54.76%		
Bambrook and Sproul (2012)	PVT air collector	Electrical efficiency 10.6–12.2% Thermal efficiency 55–60% Overall efficiency 28–55%		
Kumar and Rosen (2011)	PVT air collector (double pass with fins)	Electrical efficiency 10.5% Thermal efficiency 15.5%	66 °C	16 °C (in cell temperature)
Sarhaddi et al. (2010a,b)	PVT air collector	Electrical efficiency 10.01% Thermal efficiency 17.18% Overall efficiency 45%		
Tonui and Tripanagnostopoulos (2007b)	PVT air collector with fin		30% increase in thermal efficiency (with fin)	

Acknowledgement

The authors would like to acknowledge the financial support from the University Malaya Research Grant (UMRG) scheme (Project No: RP016A-15SUS) to carry out this research.

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